

Network Security
Network Security Project
Virtual Network Using VirtualBox with SSH Access

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1 Introduction

Secure Shell (SSH) is a widely used network protocol that enables secure communication between devices over unsecured networks. It provides encrypted remote access, ensuring confidentiality and integrity of transmitted data.

In this project, a virtual network environment was created using VirtualBox to simulate communication between multiple machines. Each virtual machine was configured with a static IP address to enable controlled connectivity within the network.

The project focuses on implementing secure remote access using SSH, applying firewall rules for access control, monitoring system logs to detect unauthorized activities, and analyzing network traffic using Wireshark. This hands-on approach demonstrates fundamental concepts of network security in a practical environment.

2 Project Objectives

The main objectives of this project are:

- Build a virtual network using VirtualBox.
- Configure static IP addresses for all virtual machines.
- Establish secure remote connections using SSH.
- Transfer files securely between machines using SCP.
- Apply firewall rules to control and restrict network access.
- Monitor system logs to detect unauthorized access attempts.
- Analyze network traffic using Wireshark.

3 Part A: Virtual Network Setup

```

ubuntu@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:5c:09:92 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 85906sec preferred_lft 85906sec
    inet6 fd17:625c:f037:2:bba4:207d:8ed1:907e/64 scope global temporary dynamic
        valid_lft 86180sec preferred_lft 14180sec
    inet6 fd17:625c:f037:2:a00:27ff:fe5c:992/64 scope global dynamic mngtmpaddr
        valid_lft 86180sec preferred_lft 14180sec
    inet6 fe80::a00:27ff:fe5c:992/64 scope link
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:1c:c2:55 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.101/24 brd 192.168.56.255 scope global noprefixroute enp0s8
        valid_lft forever preferred_lft forever
    inet 192.168.56.104/24 brd 192.168.56.255 scope global secondary dynamic noprefixroute enp0s8
        valid_lft 406sec preferred_lft 406sec
    inet6 fe80::a00:27ff:fe1c:c255/64 scope link
        valid_lft forever preferred_lft forever
ubuntu@ubuntu:~$ ip route
default via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
192.168.56.0/24 dev enp0s8 proto kernel scope link src 192.168.56.101 metric 101
192.168.56.0/24 dev enp0s8 proto kernel scope link src 192.168.56.104 metric 101
ubuntu@ubuntu:~$

```

Figure 1: IP address configuration on Ubuntu Base using the ip addr command

The `ip addr` command was used to display the network interfaces and their assigned IP addresses. The output shows that the machine has two active interfaces: one connected to the NAT network (10.0.2.15) and another connected to the Host-Only network (192.168.56.101). This confirms correct configuration for both internet access and internal communication. The Host-Only interface is used for communication between virtual machines, while the NAT interface provides internet connectivity.

```

ubuntu@ubuntu:~$ ping 192.168.56.102 #ubuntu-2
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data:
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=4.91 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=2.21 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=5.49 ms
64 bytes from 192.168.56.102: icmp_seq=4 ttl=64 time=2.50 ms
64 bytes from 192.168.56.102: icmp_seq=5 ttl=64 time=5.29 ms
64 bytes from 192.168.56.102: icmp_seq=6 ttl=64 time=2.25 ms
^X64 bytes from 192.168.56.102: icmp_seq=7 ttl=64 time=2.36 ms
^Z
[1]+  Stopped                  ping 192.168.56.102

```

Figure 2: Ping test from Ubuntu Base to ubuntu-1 (192.168.56.102)

The ping command was used to test connectivity between Ubuntu Base and ubuntu-1. The successful replies confirm that the machines are properly connected within the same network. The response time and TTL values indicate stable communication with no packet loss.

```

ubuntu@ubuntu:~$ ping 192.168.56.103
PING 192.168.56.103 (192.168.56.103) 56(84) bytes of data.
64 bytes from 192.168.56.103: icmp_seq=1 ttl=64 time=1.28 ms
64 bytes from 192.168.56.103: icmp_seq=2 ttl=64 time=0.538 ms
64 bytes from 192.168.56.103: icmp_seq=3 ttl=64 time=0.663 ms
64 bytes from 192.168.56.103: icmp_seq=4 ttl=64 time=0.505 ms
64 bytes from 192.168.56.103: icmp_seq=5 ttl=64 time=0.577 ms
64 bytes from 192.168.56.103: icmp_seq=6 ttl=64 time=0.526 ms
64 bytes from 192.168.56.103: icmp_seq=7 ttl=64 time=0.452 ms
^C
-- 192.168.56.103 ping statistics --
7 packets transmitted, 7 received, 0% packet loss, time 6145ms
rtt min/avg/max/mdev = 0.452/0.649/1.283/0.265 ms
ubuntu@ubuntu:~$

```

Figure 3: Ping test from Ubuntu Base to ubuntu-2 (192.168.56.103)

The ping command was executed to verify connectivity between Ubuntu Base and ubuntu-2. The results show successful packet transmission with zero packet loss, indicating reliable communication between the machines. The consistent response times confirm that the network is functioning correctly. This further confirms that all virtual machines are correctly connected within the Host-Only network.

```

ubuntu@ubuntu-3:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:5c:09:92 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 86351sec preferred_lft 86351sec
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:9a:52:b0 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.103/24 brd 192.168.56.255 scope global noprefixroute enp0s8
        valid_lft forever preferred_lft forever
ubuntu@ubuntu-3:~$ ip route
default via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
192.168.56.0/24 dev enp0s8 proto kernel scope link src 192.168.56.103 metric 101

```

Figure 4: IP and routing configuration on ubuntu-2 using ip addr and ip route commands

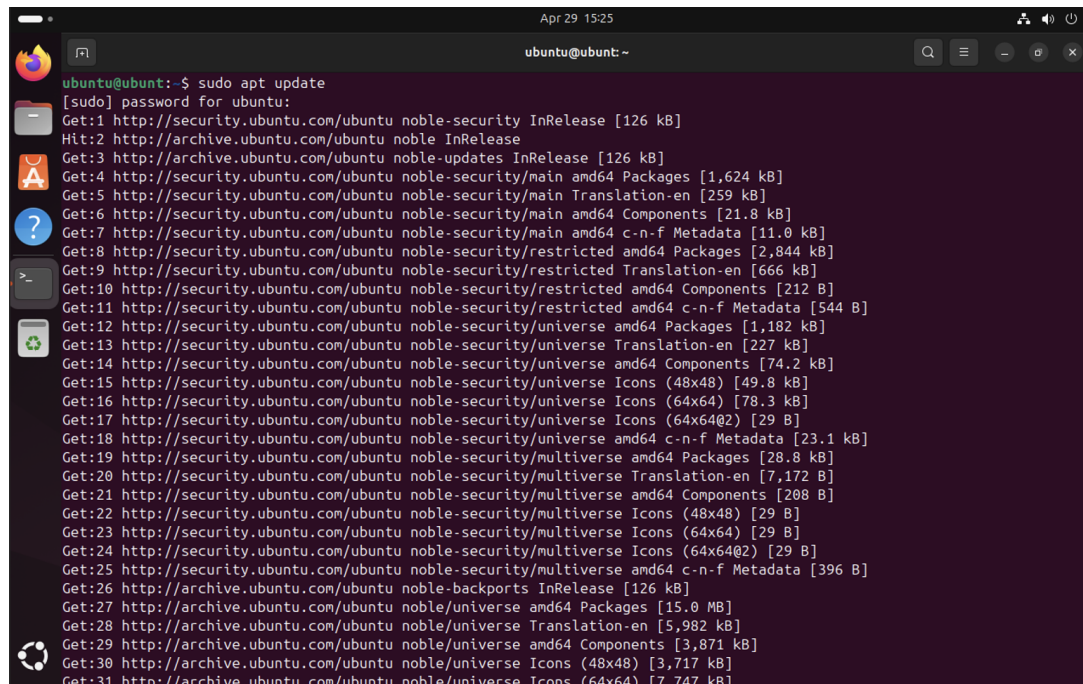
The `ip addr` and `ip route` commands were used to display the network interfaces and routing table on ubuntu-2. The output shows that the machine is assigned the IP address 192.168.56.103 on the Host-Only network and 10.0.2.15 on the NAT network. The routing table confirms that the default gateway is configured through the NAT interface, enabling internet access, while the Host-Only network allows communication with other virtual machines.

```
ubuntu@ubuntu-2:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:5c:09:92 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 86351sec preferred_lft 86351sec
    inet6 fd17:625c:f037:2:815f:ced6:77c5:5609/64 scope global temporary dynamic
        valid_lft 86352sec preferred_lft 14352sec
    inet6 fd17:625c:f037:2:a00:27ff:fe5c:992/64 scope global dynamic mngtmpaddr
        valid_lft 86352sec preferred_lft 14352sec
    inet6 fe80::a00:27ff:fe5c:992/64 scope link
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:de:72:cb brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.102/24 brd 192.168.56.255 scope global noprefixroute enp0s8
        valid_lft forever preferred_lft forever
    inet 192.168.56.103/24 brd 192.168.56.255 scope global secondary dynamic noprefixroute enp0s8
        valid_lft 551sec preferred_lft 551sec
    inet6 fe80::a00:27ff:fede:72cb/64 scope link
        valid_lft forever preferred_lft forever
ubuntu@ubuntu-2:~$ ip route
default via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
192.168.56.0/24 dev enp0s8 proto kernel scope link src 192.168.56.102 metric 101
192.168.56.0/24 dev enp0s8 proto kernel scope link src 192.168.56.103 metric 101
ubuntu@ubuntu-2:~$
```

Figure 5: IP and routing configuration on ubuntu-1 using `ip addr` and `ip route` commands

The `ip addr` and `ip route` commands were used to examine the network configuration on ubuntu-1. The output indicates that the machine is assigned the IP address 192.168.56.102 on the Host-Only network and 10.0.2.15 on the NAT network. The routing table shows that the default gateway is configured through the NAT interface, allowing internet access, while the Host-Only interface enables communication with other virtual machines in the network. This confirms proper network setup and consistent configuration across all virtual machines.

4 Part B: Secure Remote Access (SSH)



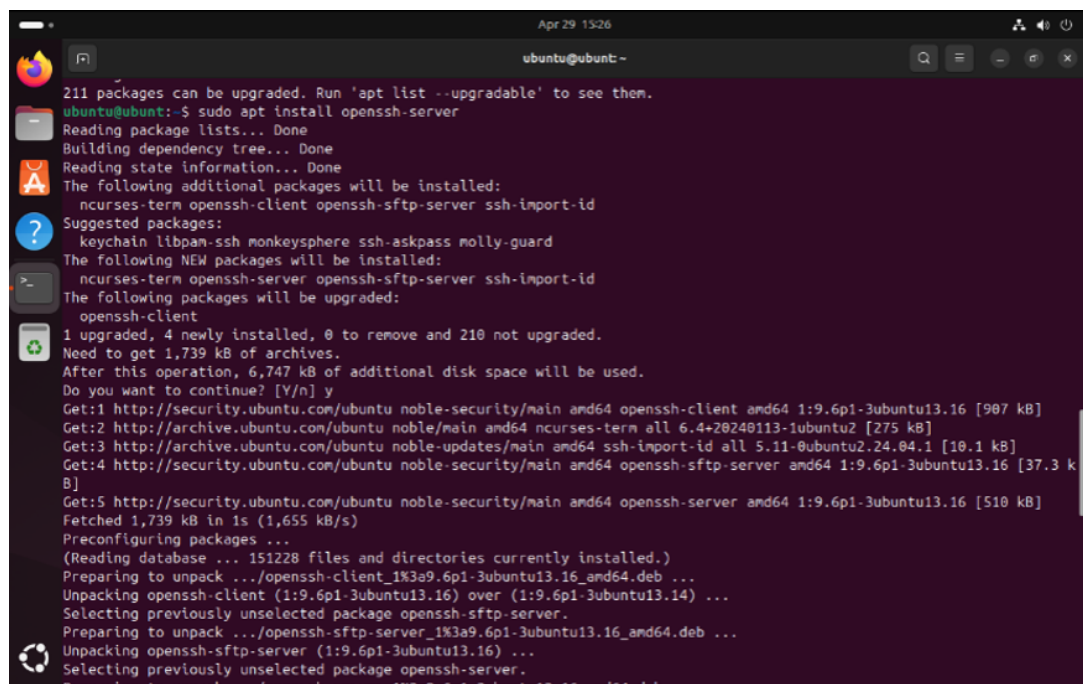
```

ubuntu@ubuntu:~$ sudo apt update
[sudo] password for ubuntu:
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,624 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [259 kB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.8 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [11.0 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2,844 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [666 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:11 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 c-n-f Metadata [544 B]
Get:12 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1,182 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [227 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.2 kB]
Get:15 http://security.ubuntu.com/ubuntu noble-security/universe Icons (48x48) [49.8 kB]
Get:16 http://security.ubuntu.com/ubuntu noble-security/universe Icons (64x64) [78.3 kB]
Get:17 http://security.ubuntu.com/ubuntu noble-security/universe Icons (64x64@2) [29 B]
Get:18 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [23.1 kB]
Get:19 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [28.8 kB]
Get:20 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [7,172 B]
Get:21 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [208 B]
Get:22 http://security.ubuntu.com/ubuntu noble-security/multiverse Icons (48x48) [29 B]
Get:23 http://security.ubuntu.com/ubuntu noble-security/multiverse Icons (64x64) [29 B]
Get:24 http://security.ubuntu.com/ubuntu noble-security/multiverse Icons (64x64@2) [29 B]
Get:25 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 c-n-f Metadata [396 B]
Get:26 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:27 http://archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:28 http://archive.ubuntu.com/ubuntu noble/universe Translation-en [5,982 kB]
Get:29 http://archive.ubuntu.com/ubuntu noble/universe amd64 Components [3,871 kB]
Get:30 http://archive.ubuntu.com/ubuntu noble/universe Icons (48x48) [3,717 kB]
Get:31 http://archive.ubuntu.com/ubuntu noble/universe Icons (64x64) [7,747 kB]

```

Figure 6: Updating package lists on Ubuntu Base

The system package list was updated using the `apt update` command to ensure that the latest software versions are available before installing the SSH service.



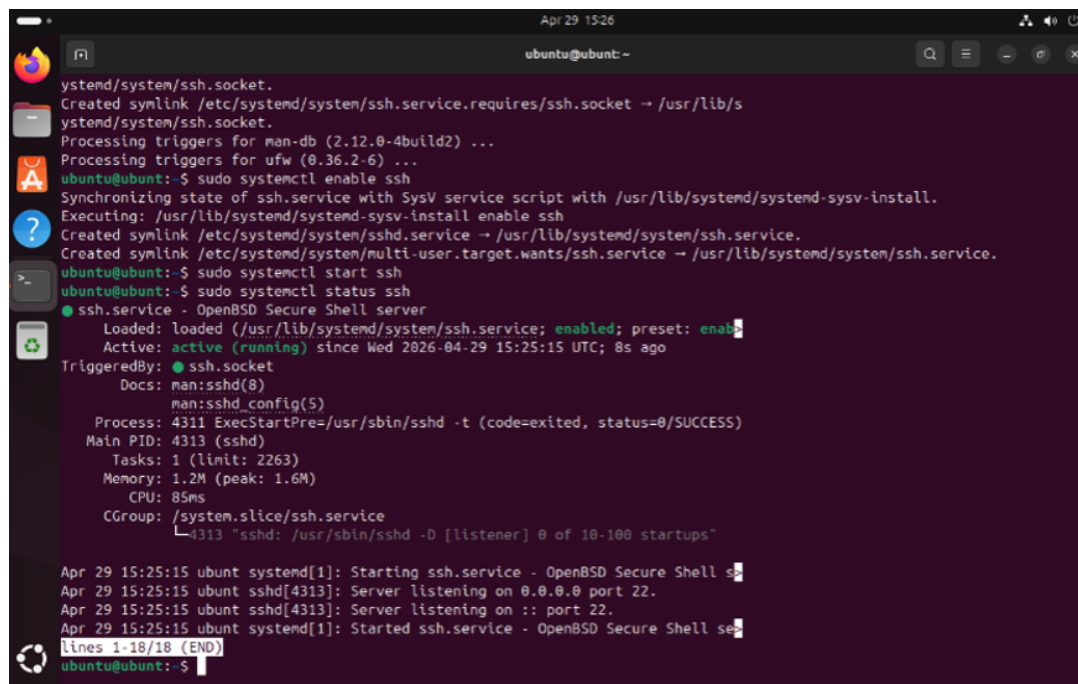
```

211 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ubuntu:~$ sudo apt install openssh-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  ncurses-term openssh-client openssh-sftp-server ssh-import-id
Suggested packages:
  keychain libpam-ssh monkeysphere ssh-askpass molly-guard
The following NEW packages will be installed:
  ncurses-term openssh-server openssh-sftp-server ssh-import-id
The following packages will be upgraded:
  openssh-client
1 upgraded, 4 newly installed, 0 to remove and 210 not upgraded.
Need to get 1,739 kB of archives.
After this operation, 6,747 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://security.ubuntu.com/ubuntu noble-security/main amd64 openssh-client amd64 1:9.6p1-3ubuntu13.16 [907 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 ncurses-term all 6.4+20240113-1ubuntu2 [275 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 ssh-import-id all 5.11-0ubuntu2.24.04.1 [10.1 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/main amd64 openssh-sftp-server amd64 1:9.6p1-3ubuntu13.16 [37.3 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 openssh-server amd64 1:9.6p1-3ubuntu13.16 [510 kB]
Fetched 1,739 kB in 1s (1,655 kB/s)
Preconfiguring packages ...
(Reading database ... 151228 files and directories currently installed.)
Preparing to unpack .../openssh-client_1:9.6p1-3ubuntu13.16_amd64.deb ...
Unpacking openssh-client (1:9.6p1-3ubuntu13.16) over (1:9.6p1-3ubuntu13.14) ...
Selecting previously unselected package openssh-sftp-server.
Preparing to unpack .../openssh-sftp-server_1:9.6p1-3ubuntu13.16_amd64.deb ...
Unpacking openssh-sftp-server (1:9.6p1-3ubuntu13.16) ...
Selecting previously unselected package openssh-server.
Preparing to unpack .../openssh-server_1:9.6p1-3ubuntu13.16_amd64.deb ...
Unpacking openssh-server (1:9.6p1-3ubuntu13.16) ...
Setting up openssh-client (1:9.6p1-3ubuntu13.16) ...
Setting up ncurses-term (6.4+20240113-1ubuntu2) ...
Setting up ssh-import-id (5.11-0ubuntu2.24.04.1) ...
Setting up openssh-sftp-server (1:9.6p1-3ubuntu13.16) ...
Setting up openssh-server (1:9.6p1-3ubuntu13.16) ...

```

Figure 7: Installing OpenSSH server on Ubuntu Base

The OpenSSH server was installed on the Ubuntu Base machine using the appropriate command. This step enables secure remote access capabilities between virtual machines.

A terminal window on an Ubuntu system showing the steps to enable and start the SSH service. The user runs 'sudo systemctl enable ssh' to enable the service at boot, followed by 'sudo systemctl start ssh' to start it immediately. Then, 'sudo systemctl status ssh' is run to check the service status. The output shows the service is 'loaded' and 'active (running)'. Below this, detailed status information is shown, including the process ID (4311), main PID (4313), and resource usage. At the bottom, system logs show the service starting and listening on port 22.

```
Apr 29 15:26
ubuntu@ubuntu: ~
systemd/system/ssh.socket.
Created symlink /etc/systemd/system/ssh.service.requires/ssh.socket → /usr/lib/s
systemd/system/ssh.socket.
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for ufw (0.36.2-6) ...
ubuntu@ubuntu:~$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
Created symlink /etc/systemd/system/ssh.service → /usr/lib/systemd/system/ssh.service.
Created symlink /etc/systemd/system/multi-user.target.wants/ssh.service → /usr/lib/systemd/system/ssh.service.
ubuntu@ubuntu:~$ sudo systemctl start ssh
ubuntu@ubuntu:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enab
   Active: active (running) since Wed 2026-04-29 15:25:15 UTC; 8s ago
   TriggeredBy: ● ssh.socket
     Docs: man:ssh(8)
           man:ssh_config(5)
   Process: 4311 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 4313 (sshd)
     Tasks: 1 (limit: 2263)
    Memory: 1.2M (peak: 1.6M)
       CPU: 85ms
   CGroup: /system.slice/ssh.service
           └─4313 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Apr 29 15:25:15 ubuntu systemd[1]: Starting ssh.service - OpenBSD Secure Shell s
Apr 29 15:25:15 ubuntu sshd[4313]: Server listening on 0.0.0.0 port 22.
Apr 29 15:25:15 ubuntu sshd[4313]: Server listening on :: port 22.
Apr 29 15:25:15 ubuntu systemd[1]: Started ssh.service - OpenBSD Secure Shell se
lines 1-18/18 (END)
ubuntu@ubuntu:~$
```

Figure 8: Enabling and verifying SSH service status

The SSH service was enabled to start automatically and then started manually. The service status confirms that SSH is active and running, allowing secure communication between the virtual machines.

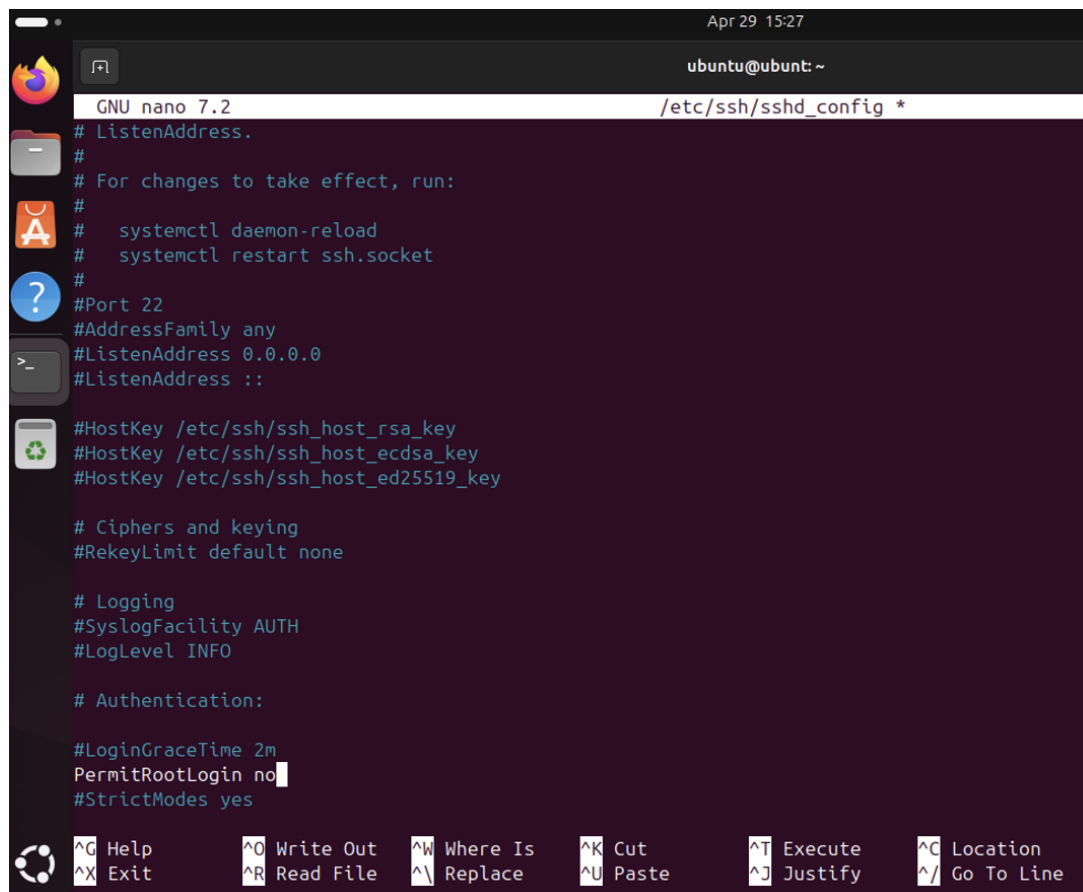


Figure 9: Editing SSH configuration file to disable root login

The SSH configuration file (`sshd.config`) was modified to improve security by disabling root login. This prevents direct access to the system using the root account, reducing the risk of unauthorized access.

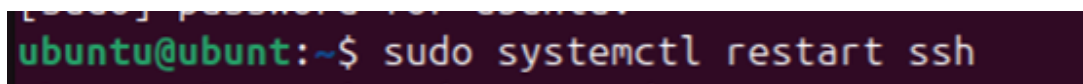
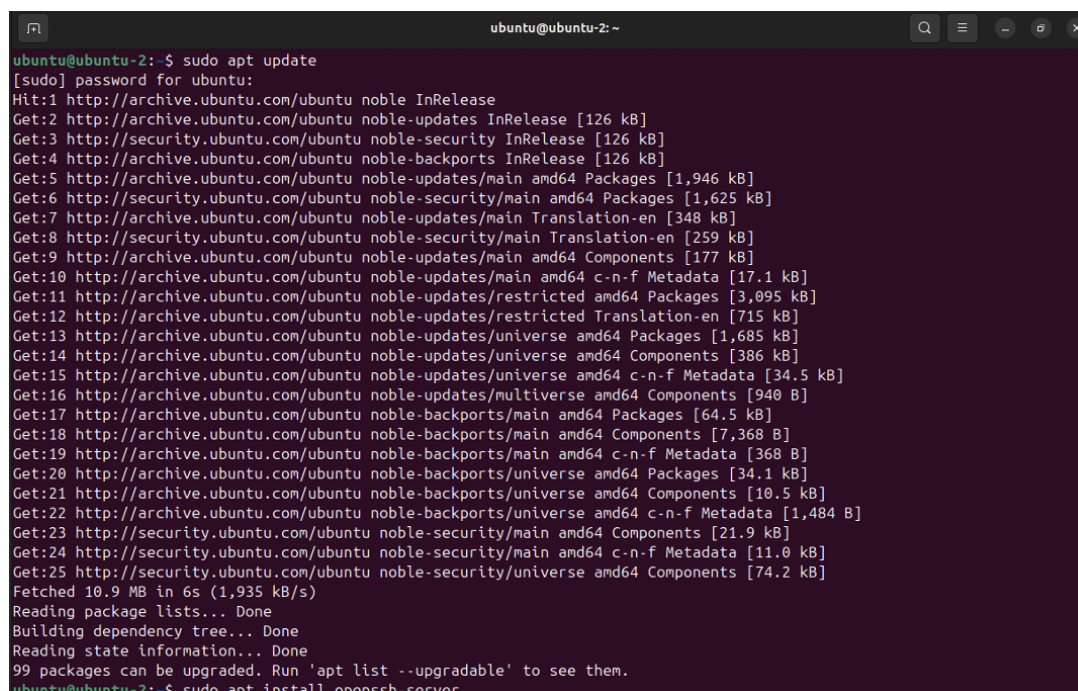


Figure 10: Restarting SSH service after configuration changes

After modifying the SSH configuration, the SSH service was restarted to apply the changes. This ensures that the updated security settings take effect immediately.



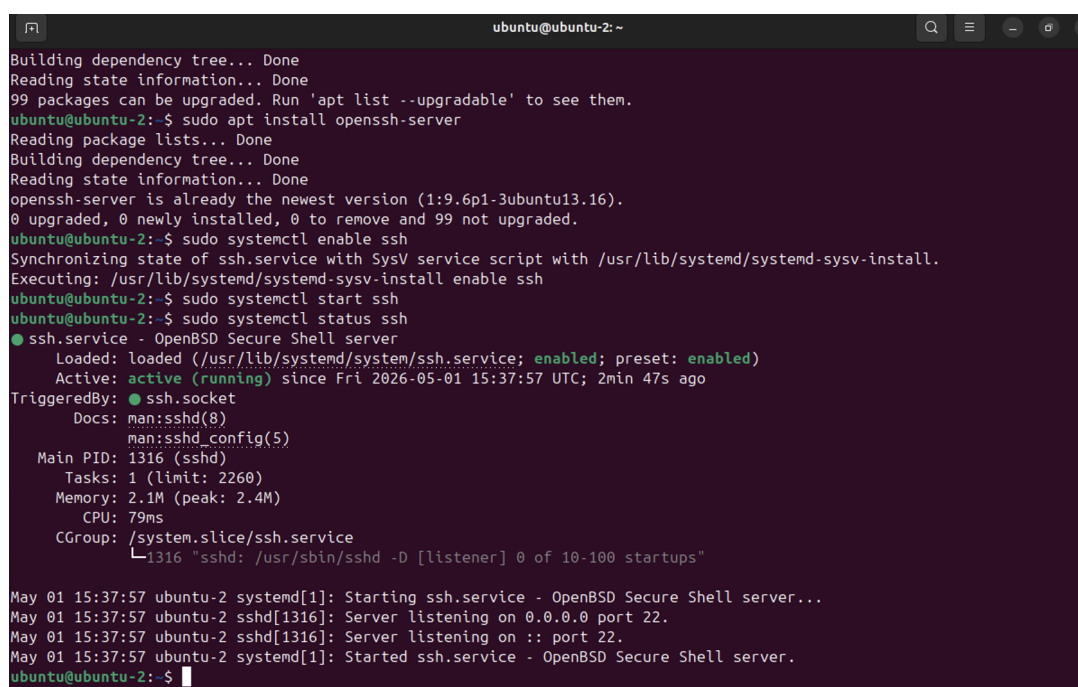
```

ubuntu@ubuntu-2: ~
ubuntu@ubuntu-2:~$ sudo apt update
[sudo] password for ubuntu:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,946 kB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,625 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [348 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [259 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [177 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [17.1 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [3,095 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [715 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1,685 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [34.5 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:17 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages [64.5 kB]
Get:18 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7,368 B]
Get:19 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 c-n-f Metadata [368 B]
Get:20 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [34.1 kB]
Get:21 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [10.5 kB]
Get:22 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 c-n-f Metadata [1,484 B]
Get:23 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.9 kB]
Get:24 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [11.0 kB]
Get:25 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.2 kB]
Fetched 10.9 MB in 6s (1,935 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
99 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ubuntu-2:~$ sudo apt install openssh-server

```

Figure 11: Updating system packages on ubuntu-1

The package list was updated on ubuntu-1 using the `apt update` command to ensure that all required packages are up to date before installing the SSH service.



```

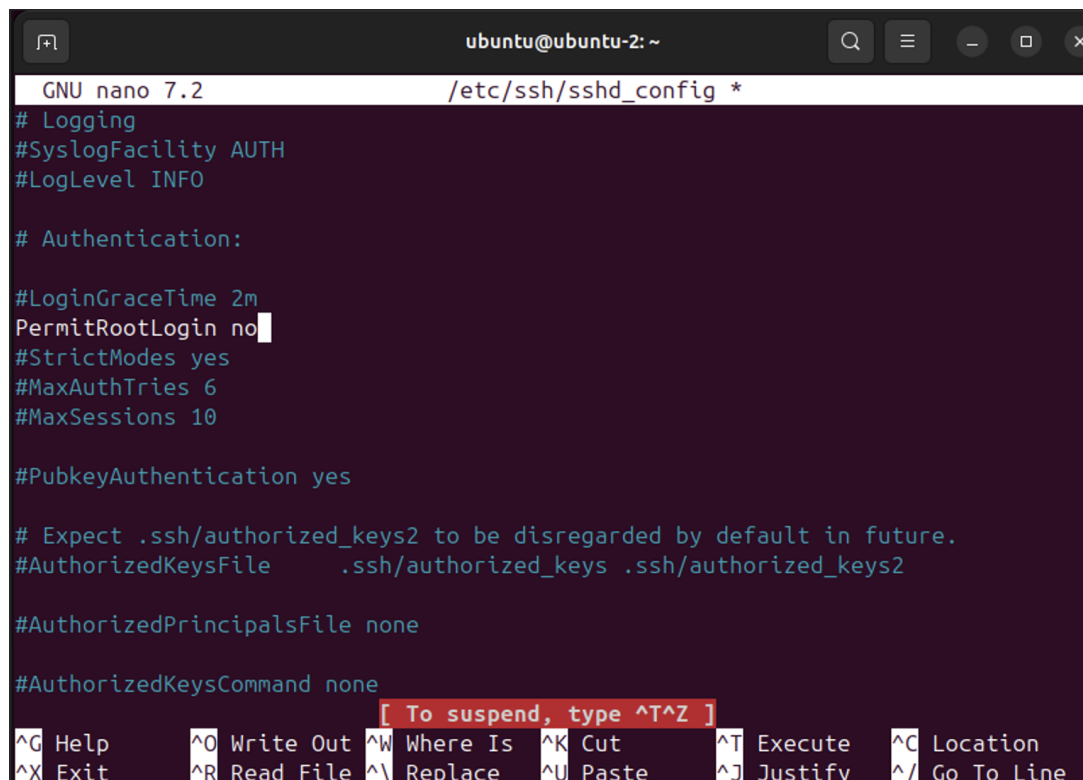
ubuntu@ubuntu-2: ~
Building dependency tree... Done
Reading state information... Done
99 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ubuntu-2:~$ sudo apt install openssh-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
openssh-server is already the newest version (1:9.6p1-3ubuntu13.16).
0 upgraded, 0 newly installed, 0 to remove and 99 not upgraded.
ubuntu@ubuntu-2:~$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
ubuntu@ubuntu-2:~$ sudo systemctl start ssh
ubuntu@ubuntu-2:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-05-01 15:37:57 UTC; 2min 47s ago
     TriggeredBy: ● ssh.socket
       Docs: man:sshd(8)
            man:sshd_config(5)
    Main PID: 1316 (sshd)
      Tasks: 1 (limit: 2260)
     Memory: 2.1M (peak: 2.4M)
        CPU: 79ms
       CGroup: /system.slice/ssh.service
              └─1316 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

May 01 15:37:57 ubuntu-2 systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
May 01 15:37:57 ubuntu-2 sshd[1316]: Server listening on 0.0.0.0 port 22.
May 01 15:37:57 ubuntu-2 sshd[1316]: Server listening on :: port 22.
May 01 15:37:57 ubuntu-2 systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
ubuntu@ubuntu-2:~$

```

Figure 12: Installing and verifying SSH service on ubuntu-1

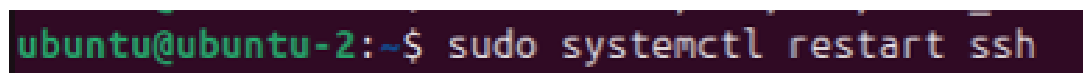
The OpenSSH server was installed on ubuntu-1, and the SSH service was started and verified. The status output confirms that the service is active and running, allowing secure remote connections.



```
ubuntu@ubuntu-2: ~  
GNU nano 7.2 /etc/ssh/sshd_config *  
# Logging  
#SyslogFacility AUTH  
#LogLevel INFO  
  
# Authentication:  
  
#LoginGraceTime 2m  
PermitRootLogin no  
#StrictModes yes  
#MaxAuthTries 6  
#MaxSessions 10  
  
#PubkeyAuthentication yes  
  
# Expect .ssh/authorized_keys2 to be disregarded by default in future.  
#AuthorizedKeysFile .ssh/authorized_keys .ssh/authorized_keys2  
  
#AuthorizedPrincipalsFile none  
  
#AuthorizedKeysCommand none  
[ To suspend, type ^T^Z ]  
^G Help      ^O Write Out ^W Where Is  ^K Cut      ^T Execute  ^C Location  
^X Exit      ^R Read File ^\ Replace   ^U Paste    ^J Justify  ^_ Go To Line
```

Figure 13: Modifying SSH configuration to disable root login on ubuntu-1

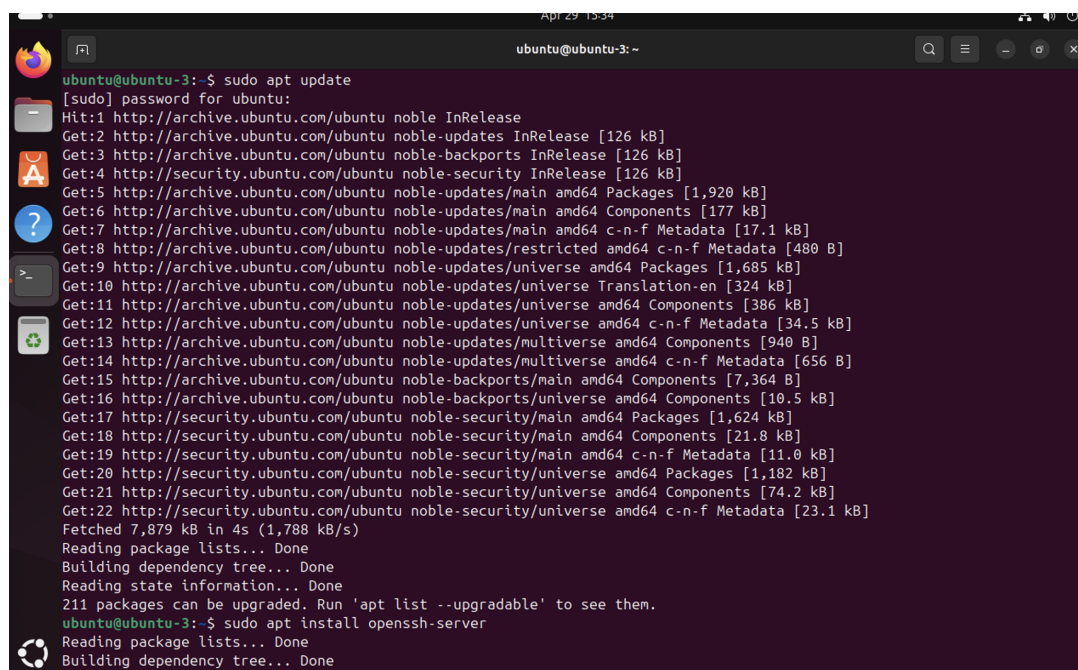
The SSH configuration file (`sshd_config`) on ubuntu-1 was edited to disable root login. This security measure helps prevent direct administrative access and reduces the risk of unauthorized login attempts.



```
ubuntu@ubuntu-2:~$ sudo systemctl restart ssh
```

Figure 14: Restarting SSH service on ubuntu-1 to apply configuration changes

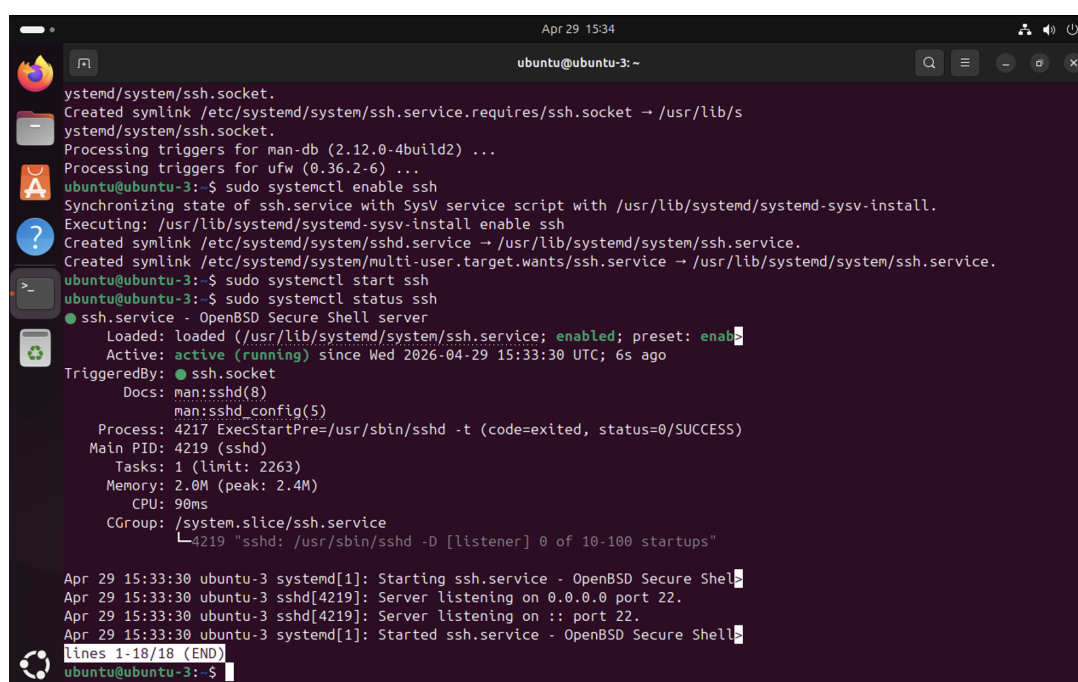
After updating the SSH configuration, the SSH service was restarted to ensure that the new settings are applied. This step activates the security configuration changes on the system.

A terminal window on an Ubuntu system. The user runs 'sudo apt update', which shows a list of updates from various sources (archive.ubuntu.com, security.ubuntu.com) including InRelease files, Packages, Components, and Metadata. It reports that 211 packages can be upgraded. Then, the user runs 'sudo apt install openssh-server', which shows the package being installed from the security.ubuntu.com source.

```
ubuntu@ubuntu-3: ~$ sudo apt update
[sudo] password for ubuntu:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,920 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [177 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [17.1 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 c-n-f Metadata [480 B]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1,685 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [324 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [34.5 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 c-n-f Metadata [656 B]
Get:15 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7,364 B]
Get:16 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [10.5 kB]
Get:17 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,624 kB]
Get:18 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.8 kB]
Get:19 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [11.0 kB]
Get:20 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1,182 kB]
Get:21 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.2 kB]
Get:22 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [23.1 kB]
Fetched 7,879 kB in 4s (1,788 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
211 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ubuntu-3: ~$ sudo apt install openssh-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

Figure 15: Updating packages and installing OpenSSH server on ubuntu-2

The system packages were updated on ubuntu-2, followed by the installation of the OpenSSH server. This prepares the machine to support secure remote access.

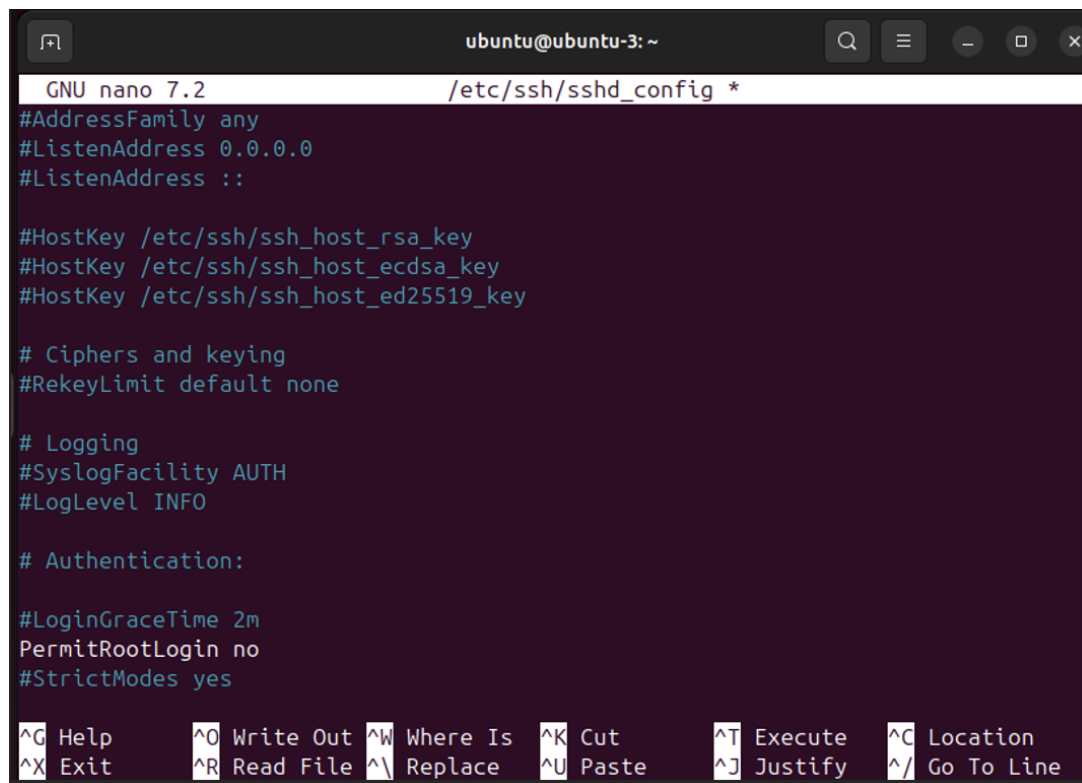
A terminal window showing the process of starting the SSH service. The user runs 'sudo systemctl enable ssh', which creates symlinks and processes triggers. Then, the user runs 'sudo systemctl start ssh', which starts the service. Finally, the user runs 'sudo systemctl status ssh', which shows the service is active (running) and listening on port 22. The status output includes details about the service's configuration, process, and resource usage.

```
systemd/systemd/ssh.socket.
Created symlink /etc/systemd/system/ssh.service.requires/ssh.socket → /usr/lib/s
systemd/systemd/ssh.socket.
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for ufw (0.36.2-6) ...
ubuntu@ubuntu-3: ~$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
Created symlink /etc/systemd/system/ssh.service → /usr/lib/systemd/system/ssh.service.
Created symlink /etc/systemd/system/multi-user.target.wants/ssh.service → /usr/lib/systemd/system/ssh.service.
ubuntu@ubuntu-3: ~$ sudo systemctl start ssh
ubuntu@ubuntu-3: ~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enab
   Active: active (running) since Wed 2026-04-29 15:33:30 UTC; 6s ago
   TriggeredBy: ● ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
    Process: 4217 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 4219 (sshd)
      Tasks: 1 (limit: 2263)
     Memory: 2.0M (peak: 2.4M)
        CPU: 90ms
    CGroup: /system.slice/ssh.service
            └─4219 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Apr 29 15:33:30 ubuntu-3 systemd[1]: Starting ssh.service - OpenBSD Secure Shel
Apr 29 15:33:30 ubuntu-3 sshd[4219]: Server listening on 0.0.0.0 port 22.
Apr 29 15:33:30 ubuntu-3 sshd[4219]: Server listening on :: port 22.
Apr 29 15:33:30 ubuntu-3 systemd[1]: Started ssh.service - OpenBSD Secure Shell
lines 1-18/18 (END)
ubuntu@ubuntu-3: ~$
```

Figure 16: Starting and verifying SSH service status on ubuntu-2

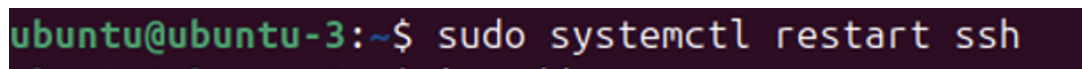
The SSH service was started and verified on ubuntu-2. The status output shows that the service is active and listening, confirming that the machine is ready to accept secure SSH connections.



```
ubuntu@ubuntu-3: ~  
GNU nano 7.2 /etc/ssh/sshd_config *  
#AddressFamily any  
#ListenAddress 0.0.0.0  
#ListenAddress ::  
  
#HostKey /etc/ssh/ssh_host_rsa_key  
#HostKey /etc/ssh/ssh_host_ecdsa_key  
#HostKey /etc/ssh/ssh_host_ed25519_key  
  
# Ciphers and keying  
#RekeyLimit default none  
  
# Logging  
#SyslogFacility AUTH  
#LogLevel INFO  
  
# Authentication:  
  
#LoginGraceTime 2m  
PermitRootLogin no  
#StrictModes yes  
  
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute  ^C Location  
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify  ^_ Go To Line
```

Figure 17: Updating SSH configuration to disable root login on ubuntu-2

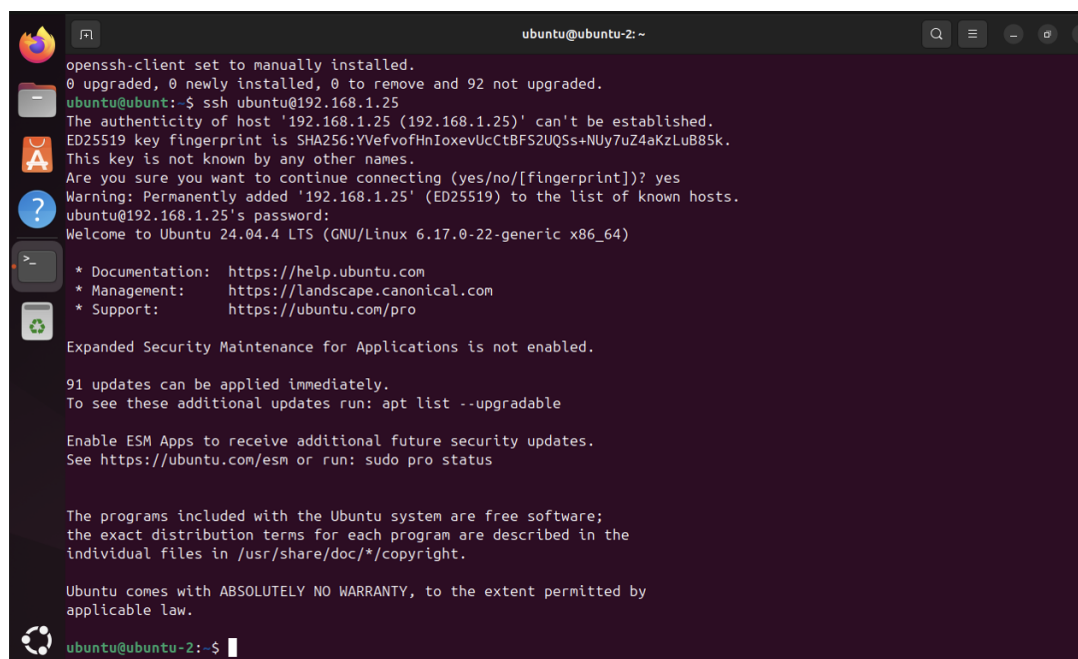
The SSH configuration file (`sshd.config`) was edited on ubuntu-2 to disable root login. This enhances system security by preventing direct administrative access and reducing potential attack vectors.



```
ubuntu@ubuntu-3:~$ sudo systemctl restart ssh
```

Figure 18: Restarting SSH service to apply security changes on ubuntu-2

The SSH service was restarted to apply the updated configuration settings. This ensures that the new security restrictions are enforced on the system.



```

ubuntu@ubuntu-2: ~
openssh-client set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 92 not upgraded.
ubuntu@ubuntu:~$ ssh ubuntu@192.168.1.25
The authenticity of host '192.168.1.25 (192.168.1.25)' can't be established.
ED25519 key fingerprint is SHA256:YVefvofHnIoxevUcCtBF52UQ5s+NUy7uZ4aKzLuB85k.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.1.25' (ED25519) to the list of known hosts.
ubuntu@192.168.1.25's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-22-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

91 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

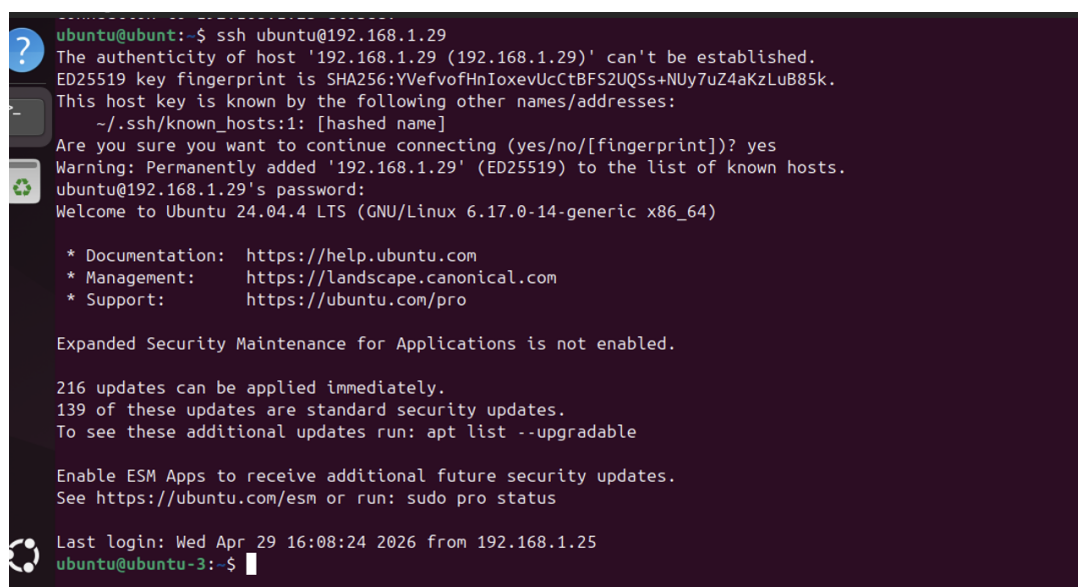
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ubuntu@ubuntu-2:~$

```

Figure 19: Establishing an SSH connection from Ubuntu Base to ubuntu-1

An SSH connection was successfully established from Ubuntu Base to ubuntu-1 using its IP address. During the first connection, the system prompted to verify the host authenticity and add the ECDSA key to the list of known hosts. After confirming, the connection was established successfully, allowing secure remote access to the target machine.



```

ubuntu@ubuntu:~$ ssh ubuntu@192.168.1.29
The authenticity of host '192.168.1.29 (192.168.1.29)' can't be established.
ED25519 key fingerprint is SHA256:YVefvofHnIoxevUcCtBF52UQ5s+NUy7uZ4aKzLuB85k.
This host key is known by the following other names/addresses:
~/.ssh/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.1.29' (ED25519) to the list of known hosts.
ubuntu@192.168.1.29's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-14-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

216 updates can be applied immediately.
139 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Wed Apr 29 16:08:24 2026 from 192.168.1.25
ubuntu@ubuntu-3:~$

```

Figure 20: Establishing an SSH connection from Ubuntu Base to ubuntu-2

An SSH connection was successfully established from Ubuntu Base to ubuntu-2 using its IP address. During the initial connection, the system prompted for host verification and added the server's key to the known hosts file. After authentication, access was granted, confirming secure communication between the machines.

```
ubuntu@ubuntu-2:~$ ssh ubuntu@192.168.1.29
ubuntu@192.168.1.29's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-14-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

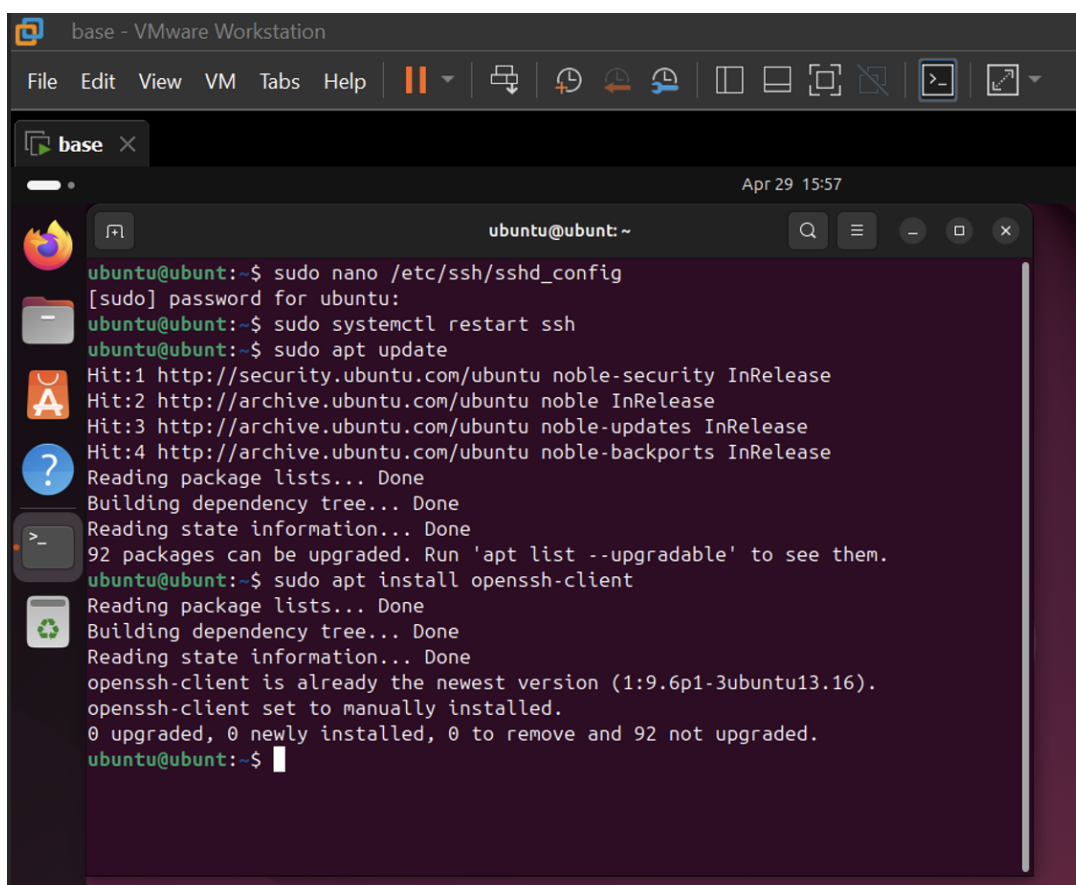
216 updates can be applied immediately.
139 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Wed Apr 29 16:14:27 2026 from 192.168.1.27
ubuntu@ubuntu-3:~$
```

Figure 21: SSH connection from ubuntu-1 to ubuntu-2

An SSH connection was successfully established from ubuntu-1 to ubuntu-2 using the target machine's IP address. The successful login confirms that secure communication is enabled not only from the base machine but also between client virtual machines within the network.



```
base - VMware Workstation
File Edit View VM Tabs Help
ubuntu@ubuntu: ~
Apr 29 15:57
ubuntu@ubuntu:~$ sudo nano /etc/ssh/sshd_config
[sudo] password for ubuntu:
ubuntu@ubuntu:~$ sudo systemctl restart ssh
ubuntu@ubuntu:~$ sudo apt update
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
92 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ubuntu:~$ sudo apt install openssh-client
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
openssh-client is already the newest version (1:9.6p1-3ubuntu13.16).
openssh-client set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 92 not upgraded.
ubuntu@ubuntu:~$
```

Figure 22: Installing SSH client on Ubuntu Base

The package list was updated on Ubuntu Base, and the OpenSSH client was installed to enable outgoing SSH connections. The output confirms that the SSH client is already installed and up to date, ensuring that the system can initiate secure remote connections to other virtual

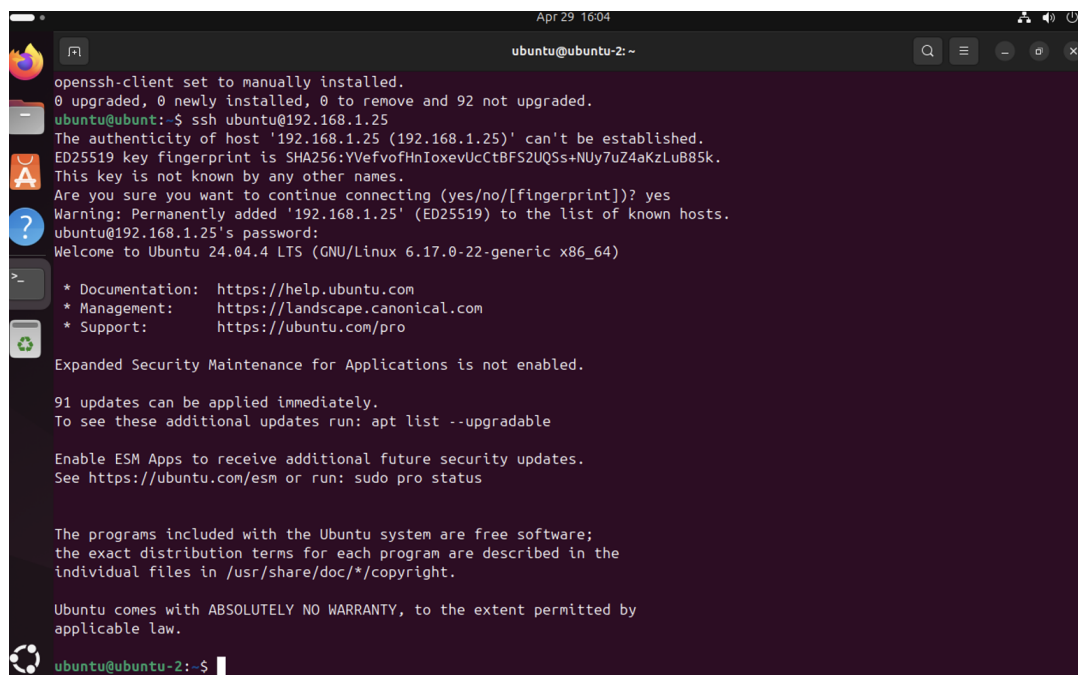
machines. This step ensures that Ubuntu Base can act as an SSH client to communicate with other machines in the network.

5 Part C: Core Tasks and Evidence

```
ubuntu@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:50:e3:71 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.1.27/24 brd 192.168.1.255 scope global dynamic noprefixroute ens33
        valid_lft 85088sec preferred_lft 85088sec
    inet6 2001:16a2:f437:1700:6bd2:cf67:19bc:b32b/64 scope global temporary dynamic
        valid_lft 86151sec preferred_lft 84900sec
    inet6 2001:16a2:f437:1700:433:5600:3b29:32af/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86151sec preferred_lft 86151sec
    inet6 fe80::215:8342:5bf8:bd7b/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: ens34: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:50:e3:7b brd ff:ff:ff:ff:ff:ff
    altname enp2s2
    inet 192.168.1.28/24 brd 192.168.1.255 scope global dynamic noprefixroute ens34
        valid_lft 85088sec preferred_lft 85088sec
    inet6 2001:16a2:f437:1700:2e01:a3b4:f57f:7011/64 scope global temporary dynamic
        valid_lft 86151sec preferred_lft 84566sec
    inet6 2001:16a2:f437:1700:a0ac:e668:acb9:3371/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86151sec preferred_lft 86151sec
    inet6 fe80::d5c4:a77a:ec44:f829/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
ubuntu@ubuntu:~$
```

Figure 23: IP address verification on Ubuntu Base using ip addr command

The `ip addr` command was used on Ubuntu Base to verify the network interfaces and their assigned IP addresses. The output shows both NAT and Host-Only interfaces, confirming that the machine is properly configured for internet access and internal network communication.



```

Apr 29 16:04
ubuntu@ubuntu-2: ~
openssh-client set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 92 not upgraded.
ubuntu@ubuntu:~$ ssh ubuntu@192.168.1.25
The authenticity of host '192.168.1.25 (192.168.1.25)' can't be established.
ED25519 key fingerprint is SHA256:YVefvofHnIoxevUcCtBFS2UQs+NUy7uZ4aKzLuB85k.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.1.25' (ED25519) to the list of known hosts.
ubuntu@192.168.1.25's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-22-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

91 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

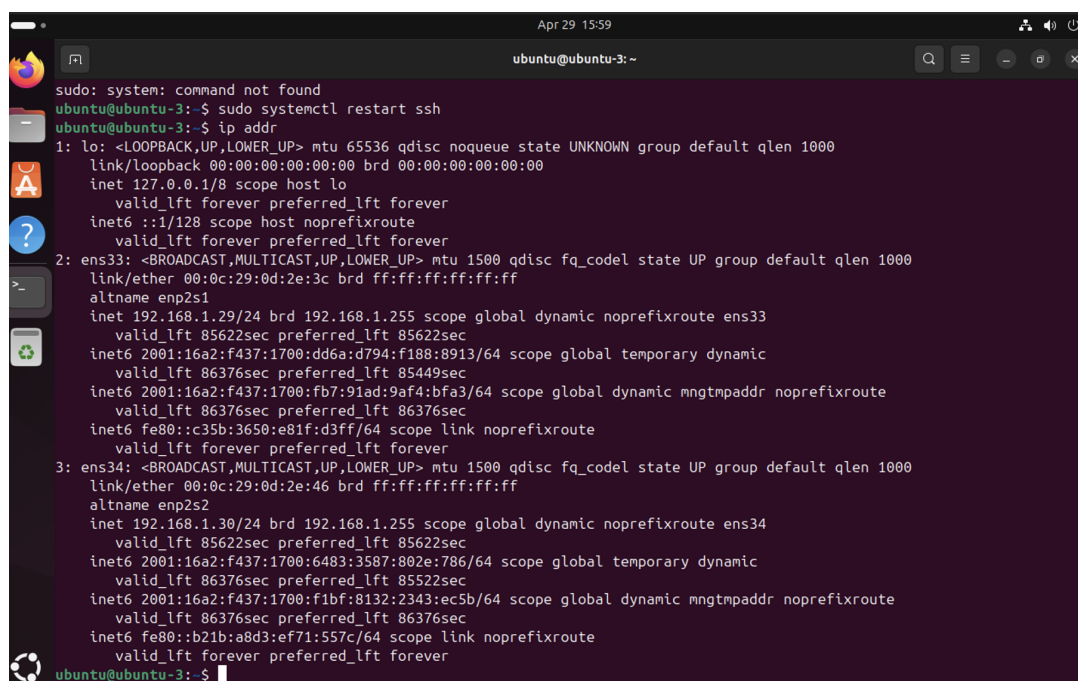
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ubuntu@ubuntu-2:~$

```

Figure 24: Displaying the network interface details and confirming the assigned IP address for the ubuntu-1 virtual machine.

An SSH connection was established to ubuntu-1 to verify network connectivity. The system prompted for host authentication and added the server key to the known hosts file. The successful login confirms that ubuntu-1 is accessible and properly connected within the network.



```

Apr 29 15:59
ubuntu@ubuntu-3: ~
sudo: system: command not found
ubuntu@ubuntu-3:~$ sudo systemctl restart ssh
ubuntu@ubuntu-3:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:0d:2e:3c brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.1.29/24 brd 192.168.1.255 scope global dynamic noprefixroute ens33
        valid_lft 85622sec preferred_lft 85622sec
    inet6 2001:16a2:f437:1700:dd6a:d794:f188:8913/64 scope global temporary dynamic
        valid_lft 86376sec preferred_lft 85449sec
    inet6 2001:16a2:f437:1700:fb7:91ad:9af4:bfa3/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86376sec preferred_lft 86376sec
    inet6 fe80::c35b:3650:e81f:d3ff/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: ens34: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:0d:2e:46 brd ff:ff:ff:ff:ff:ff
    altname enp2s2
    inet 192.168.1.30/24 brd 192.168.1.255 scope global dynamic noprefixroute ens34
        valid_lft 85622sec preferred_lft 85622sec
    inet6 2001:16a2:f437:1700:6483:3587:802e:786/64 scope global temporary dynamic
        valid_lft 86376sec preferred_lft 85522sec
    inet6 2001:16a2:f437:1700:f1bf:8132:2343:ec5b/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86376sec preferred_lft 86376sec
    inet6 fe80::b21b:a8d3:ef71:557c/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
ubuntu@ubuntu-3:~$

```

Figure 25: IP address verification on ubuntu-2 using ip addr command

The `ip addr` command was executed on ubuntu-2 to verify its network configuration and assigned IP addresses. The output shows that the machine is correctly configured with both NAT and Host-Only interfaces, confirming proper connectivity within the virtual network.

```
ubuntu@ubuntu:~$ ping 192.168.1.25
PING 192.168.1.25 (192.168.1.25) 56(84) bytes of data.
64 bytes from 192.168.1.25: icmp_seq=1 ttl=64 time=1.42 ms
64 bytes from 192.168.1.25: icmp_seq=2 ttl=64 time=2.05 ms
64 bytes from 192.168.1.25: icmp_seq=3 ttl=64 time=0.756 ms
64 bytes from 192.168.1.25: icmp_seq=4 ttl=64 time=0.806 ms
64 bytes from 192.168.1.25: icmp_seq=5 ttl=64 time=1.26 ms
64 bytes from 192.168.1.25: icmp_seq=6 ttl=64 time=0.899 ms
64 bytes from 192.168.1.25: icmp_seq=7 ttl=64 time=2.44 ms
64 bytes from 192.168.1.25: icmp_seq=8 ttl=64 time=2.81 ms
64 bytes from 192.168.1.25: icmp_seq=9 ttl=64 time=1.15 ms
64 bytes from 192.168.1.25: icmp_seq=10 ttl=64 time=3.92 ms
64 bytes from 192.168.1.25: icmp_seq=11 ttl=64 time=1.10 ms
64 bytes from 192.168.1.25: icmp_seq=12 ttl=64 time=1.16 ms
^C
--- 192.168.1.25 ping statistics ---
12 packets transmitted, 12 received, 0% packet loss, time 11062ms
rtt min/avg/max/mdev = 0.756/1.645/3.917/0.928 ms
ubuntu@ubuntu:~$ ping 192.168.1.29
PING 192.168.1.29 (192.168.1.29) 56(84) bytes of data.
64 bytes from 192.168.1.29: icmp_seq=1 ttl=64 time=1.15 ms
64 bytes from 192.168.1.29: icmp_seq=2 ttl=64 time=1.86 ms
64 bytes from 192.168.1.29: icmp_seq=3 ttl=64 time=2.40 ms
64 bytes from 192.168.1.29: icmp_seq=4 ttl=64 time=4.10 ms
64 bytes from 192.168.1.29: icmp_seq=5 ttl=64 time=1.99 ms
64 bytes from 192.168.1.29: icmp_seq=6 ttl=64 time=0.856 ms
64 bytes from 192.168.1.29: icmp_seq=7 ttl=64 time=1.30 ms
64 bytes from 192.168.1.29: icmp_seq=8 ttl=64 time=2.40 ms
^C
--- 192.168.1.29 ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 7010ms
rtt min/avg/max/mdev = 0.856/2.009/4.104/0.953 ms
ubuntu@ubuntu:~$
```

Figure 26: Ping test from Ubuntu Base to ubuntu-1 and ubuntu-2

The ping command was used on Ubuntu Base to test network connectivity with ubuntu-1 and ubuntu-2. The results show successful transmission and reception of packets with zero packet loss, confirming reliable communication between the virtual machines within the network.

```
ubuntu@ubuntu:~$ ssh ubuntu@192.168.1.25
ubuntu@192.168.1.25's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-22-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

91 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Wed Apr 29 16:03:31 2026 from 192.168.1.27
```

Figure 27: SSH connection from Ubuntu Base to ubuntu-1

An SSH connection was established from Ubuntu Base to ubuntu-1 using the target machine's IP address. The successful login confirms that secure remote access is functioning correctly and that communication between the machines is properly established.

```
ubuntu@ubuntu:~$ ssh ubuntu@192.168.1.29
The authenticity of host '192.168.1.29 (192.168.1.29)' can't be established.
ED25519 key fingerprint is SHA256:YVefvofHnIoxevUcCtBFS2UQSs+NUy7uZ4aKzLuB85k.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.1.29' (ED25519) to the list of known hosts.
ubuntu@192.168.1.29's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-14-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

216 updates can be applied immediately.
139 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Wed Apr 29 16:08:24 2026 from 192.168.1.25
ubuntu@ubuntu-3:~$
```

Figure 28: SSH connection from Ubuntu Base to ubuntu-2

An SSH connection was successfully established from Ubuntu Base to ubuntu-2 using its IP address. During the initial connection, the system verified the host authenticity and added the server key to the known hosts file. The successful authentication confirms that secure communication between the machines is functioning correctly.

```
ubuntu@ubuntu-2:~$ ssh ubuntu@192.168.1.29
ubuntu@192.168.1.29's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-14-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

216 updates can be applied immediately.
139 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Wed Apr 29 16:14:27 2026 from 192.168.1.27
ubuntu@ubuntu-3:~$
```

Figure 29: SSH connection from ubuntu-1 to ubuntu-2

An SSH connection was successfully established from ubuntu-1 to ubuntu-2 using the target machine's IP address. The successful login confirms that secure communication is enabled between the client virtual machines, demonstrating proper network configuration and connectivity. This

demonstrates that all virtual machines within the network can communicate securely with each other using SSH.

```
ubuntu@ubunt:~$ ssh ubuntu@192.168.1.25
ubuntu@192.168.1.25's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-22-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

91 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Wed Apr 29 16:03:31 2026 from 192.168.1.27
ubuntu@ubuntu-2:~$ echo "This is a test file from Ubuntu Base"> testfile.txt
ubuntu@ubuntu-2:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  snap  Templates  testfile.txt  Videos
ubuntu@ubuntu-2:~$ exit
```

Figure 30: Creating a file remotely via SSH and verifying it on ubuntu-1

A text file named `testfile.txt` was created remotely on ubuntu-1 during an active SSH session from Ubuntu Base using the `echo` command. The `ls` command was then used to verify the successful creation of the file in the directory. This demonstrates the ability to execute remote commands and manage files securely through SSH.

```
ubuntu@ubunt:~$ scp testfile.txt ubuntu@192.168.1.29:/home/ubuntu/
ubuntu@192.168.1.29's password:
testfile.txt
ubuntu@ubunt:~$
```

Figure 31: Transferring a file from Ubuntu Base to ubuntu-2 using SCP

The SCP command was used on Ubuntu Base to securely transfer the file `testfile.txt` to the home directory of ubuntu-2. The successful transfer confirms that secure file sharing between the virtual machines is functioning correctly. This demonstrates the use of encrypted protocols for secure file transfer across the network.

```
ubuntu@ubuntu-3:~$ ls /home/ubuntu/
Desktop  Documents  Downloads  Music  Pictures  Public  snap  Templates  testfile.txt  Videos
ubuntu@ubuntu-3:~$
```

Figure 32: Verify the transformation on ubuntu-2

After authentication, the file was successfully copied to the destination directory, confirming secure file transfer between the virtual machines.

```
Expanded Security Maintenance for Applications is not enabled.

91 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

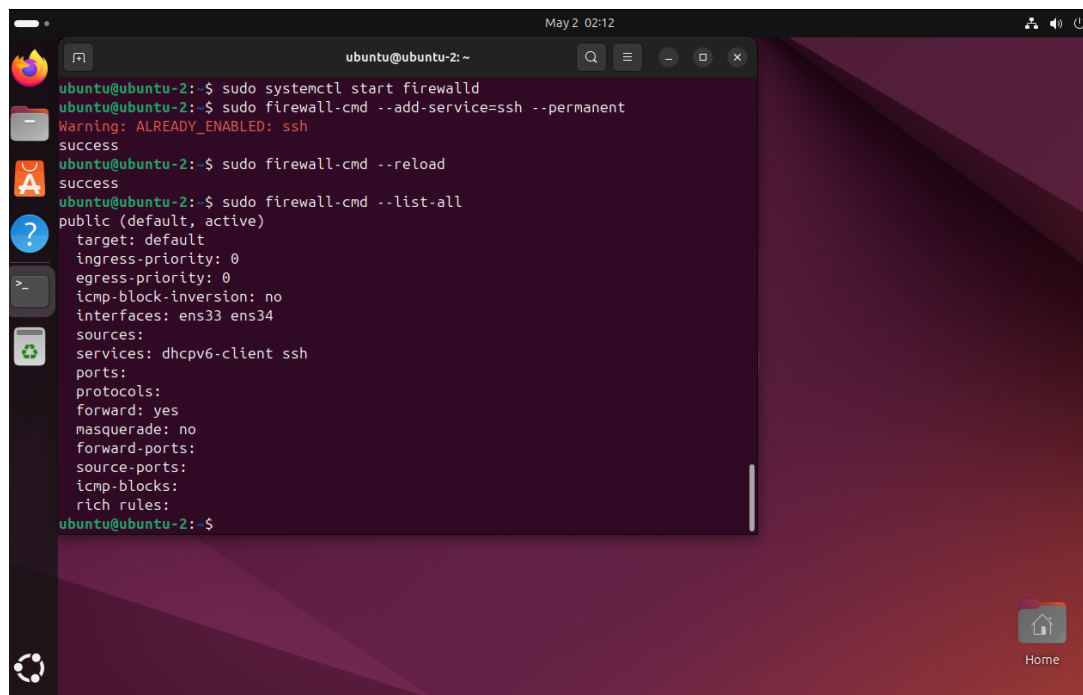
Last login: Wed Apr 29 16:24:06 2026 from 192.168.1.27
ubuntu@ubuntu-2:~$ sudo systemctl restart ssh
[sudo] password for ubuntu:
ubuntu@ubuntu-2:~$ sudo systemctl restart ssh
ubuntu@ubuntu-2:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-04-29 16:33:43 UTC; 17s ago
   TriggeredBy: ● ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
    Process: 5258 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 5259 (sshd)
      Tasks: 1 (limit: 2260)
     Memory: 1.2M (peak: 1.5M)
        CPU: 44ms
    CGroup: /system.slice/ssh.service
            └─5259 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Apr 29 16:33:43 ubuntu-2 systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Apr 29 16:33:43 ubuntu-2 sshd[5259]: Server listening on 0.0.0.0 port 22.
Apr 29 16:33:43 ubuntu-2 sshd[5259]: Server listening on :: port 22.
Apr 29 16:33:43 ubuntu-2 systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
ubuntu@ubuntu-2:~$
```

Figure 33: Restarting and verifying SSH service on ubuntu-1

The SSH service was restarted on ubuntu-1 using the `systemctl restart ssh` command during a remote SSH session. The `systemctl status ssh` command was then executed to verify that the service is active and running. The output confirms that the SSH service restarted successfully and is functioning correctly.

6 Part D: Firewall Host-Based Access Control (firewall-d)

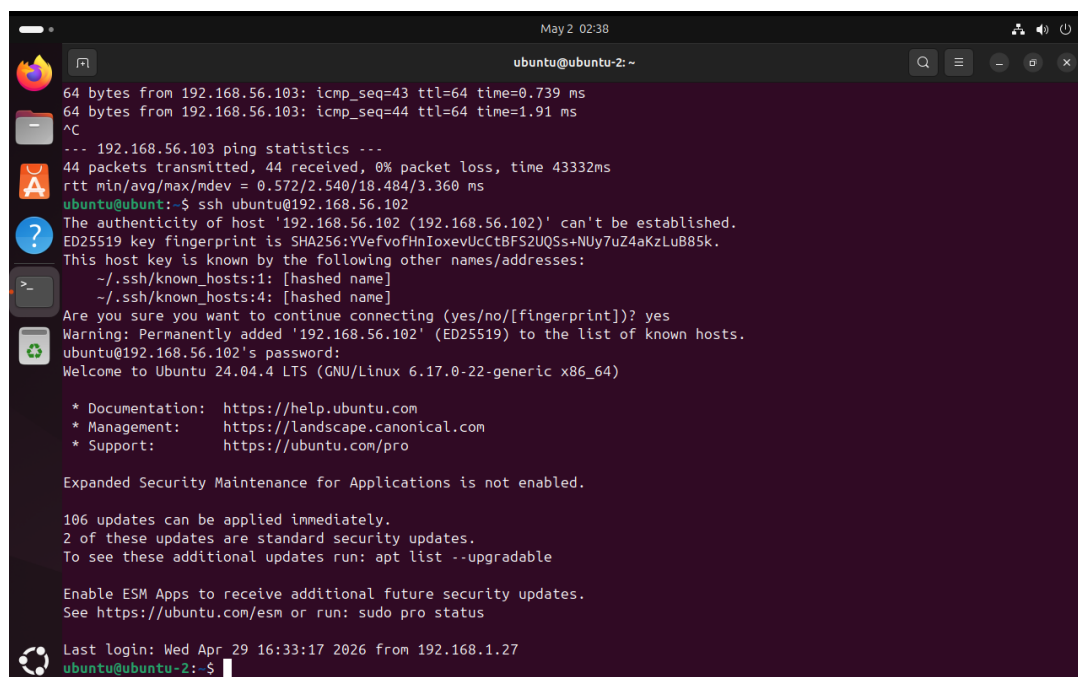
A screenshot of a terminal window on an Ubuntu desktop. The terminal shows the following commands and output:

```
ubuntu@ubuntu-2:~$ sudo systemctl start firewalld
ubuntu@ubuntu-2:~$ sudo firewall-cmd --add-service=ssh --permanent
Warning: ALREADY_ENABLED: ssh
success
ubuntu@ubuntu-2:~$ sudo firewall-cmd --reload
success
ubuntu@ubuntu-2:~$ sudo firewall-cmd --list-all
public (default, active)
target: default
ingress-priority: 0
egress-priority: 0
icmp-block-inversion: no
interfaces: ens33 ens34
sources:
services: dhcpv6-client ssh
ports:
protocols:
forward: yes
masquerade: no
forward-ports:
source-ports:
icmp-blocks:
rich rules:
ubuntu@ubuntu-2:~$
```

The terminal window is titled 'ubuntu@ubuntu-2:~' and has a search icon, a menu icon, and window control buttons. The desktop background is a dark purple gradient. A 'Home' icon is visible in the bottom right corner.

Figure 34: Configuring and verifying firewall rules on Ubuntu

The firewall service was started using `systemctl`, and the SSH service was added as an allowed rule to enable secure remote access. The firewall configuration was then reloaded to apply the changes. Finally, the `firewall-cmd --list-all` command was used to verify the active firewall rules, confirming that SSH traffic is permitted.



```

May 2 02:38
ubuntu@ubuntu-2: ~
64 bytes from 192.168.56.103: icmp_seq=43 ttl=64 time=0.739 ms
64 bytes from 192.168.56.103: icmp_seq=44 ttl=64 time=1.91 ms
^C
--- 192.168.56.103 ping statistics ---
44 packets transmitted, 44 received, 0% packet loss, time 43332ms
rtt min/avg/max/mdev = 0.572/2.540/18.484/3.360 ms
ubuntu@ubuntu:~$ ssh ubuntu@192.168.56.102
The authenticity of host '192.168.56.102 (192.168.56.102)' can't be established.
ED25519 key fingerprint is SHA256:YVefvofHnIoxevUcCtBFS2UQ5s+NUy7uZ4aKzLuB85k.
This host key is known by the following other names/addresses:
~/.ssh/known_hosts:1: [hashed name]
~/.ssh/known_hosts:4: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.102' (ED25519) to the list of known hosts.
ubuntu@192.168.56.102's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-22-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

106 updates can be applied immediately.
2 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

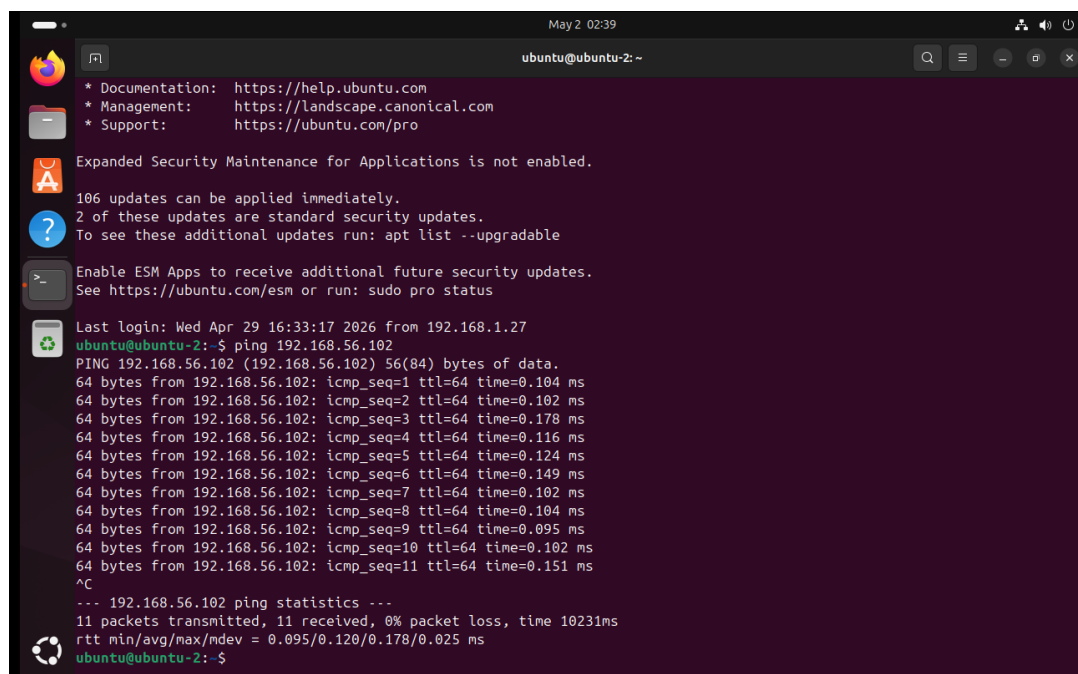
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Wed Apr 29 16:33:17 2026 from 192.168.1.27
ubuntu@ubuntu-2:~$

```

Figure 35: Verifying SSH connectivity and successful login

The SSH connection to the target machine was verified by initiating a remote login session using its IP address. During the first connection, the system confirmed the host authenticity and added it to the known hosts file. The successful login demonstrates that secure communication between the machines is properly established.



```

May 2 02:39
ubuntu@ubuntu-2: ~
 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

106 updates can be applied immediately.
2 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

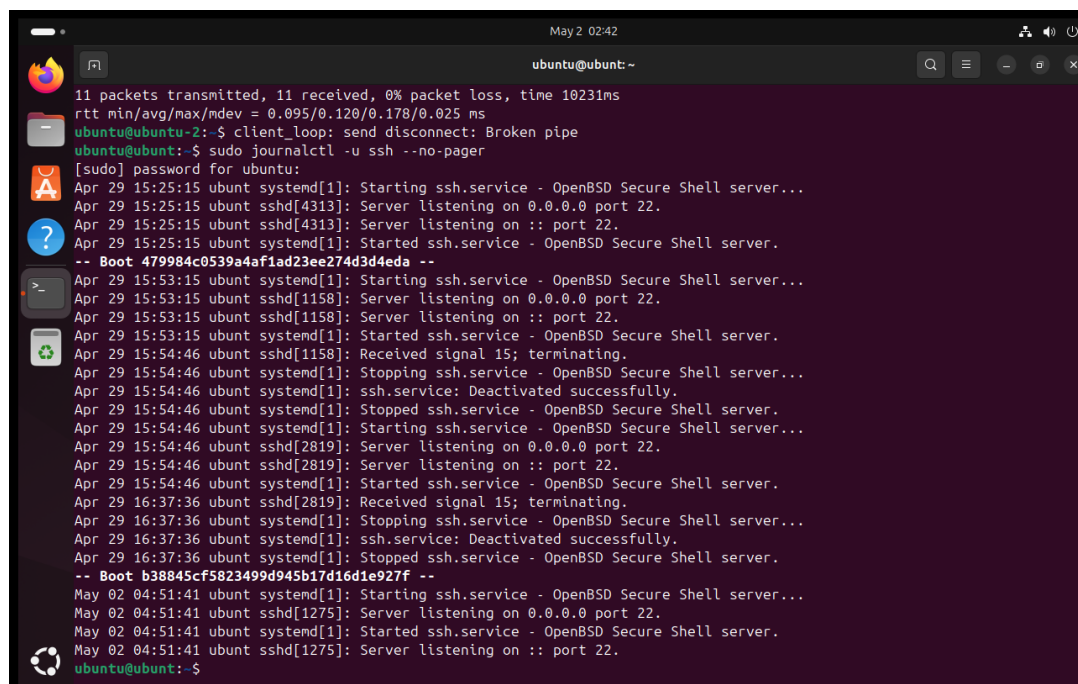
Last login: Wed Apr 29 16:33:17 2026 from 192.168.1.27
ubuntu@ubuntu-2:~$ ping 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data.
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=0.104 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=0.102 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=0.178 ms
64 bytes from 192.168.56.102: icmp_seq=4 ttl=64 time=0.116 ms
64 bytes from 192.168.56.102: icmp_seq=5 ttl=64 time=0.124 ms
64 bytes from 192.168.56.102: icmp_seq=6 ttl=64 time=0.149 ms
64 bytes from 192.168.56.102: icmp_seq=7 ttl=64 time=0.102 ms
64 bytes from 192.168.56.102: icmp_seq=8 ttl=64 time=0.104 ms
64 bytes from 192.168.56.102: icmp_seq=9 ttl=64 time=0.095 ms
64 bytes from 192.168.56.102: icmp_seq=10 ttl=64 time=0.102 ms
64 bytes from 192.168.56.102: icmp_seq=11 ttl=64 time=0.151 ms
^C
--- 192.168.56.102 ping statistics ---
11 packets transmitted, 11 received, 0% packet loss, time 10231ms
rtt min/avg/max/mdev = 0.095/0.120/0.178/0.025 ms
ubuntu@ubuntu-2:~$

```

Figure 36: Verifying network connectivity using ping

Network connectivity was verified by sending ICMP echo requests using the `ping` command to the target machine. The successful replies with no packet loss confirm stable communication and proper network configuration between the virtual machines.

7 Part E: Log Monitoring



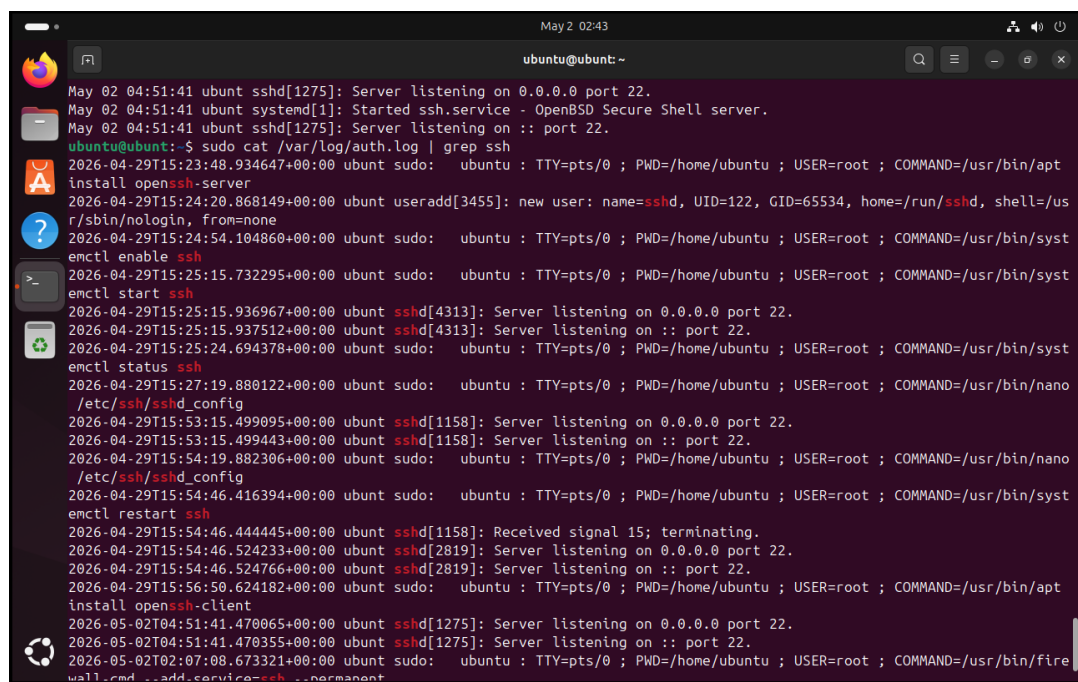
```

May 2 02:42
ubuntu@ubuntu: ~
11 packets transmitted, 11 received, 0% packet loss, time 10231ms
rtt min/avg/max/mdev = 0.095/0.120/0.178/0.025 ms
ubuntu@ubuntu-2:~$ client_loop: send disconnect: Broken pipe
ubuntu@ubuntu:~$ sudo journalctl -u ssh --no-pager
[sudo] password for ubuntu:
Apr 29 15:25:15 ubuntu systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Apr 29 15:25:15 ubuntu sshd[4313]: Server listening on 0.0.0.0 port 22.
Apr 29 15:25:15 ubuntu sshd[4313]: Server listening on :: port 22.
Apr 29 15:25:15 ubuntu systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
-- Boot 479984c0539a4af1ad23ee274d3d4eda --
Apr 29 15:53:15 ubuntu systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Apr 29 15:53:15 ubuntu sshd[1158]: Server listening on 0.0.0.0 port 22.
Apr 29 15:53:15 ubuntu sshd[1158]: Server listening on :: port 22.
Apr 29 15:53:15 ubuntu systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Apr 29 15:54:46 ubuntu sshd[1158]: Received signal 15; terminating.
Apr 29 15:54:46 ubuntu systemd[1]: Stopping ssh.service - OpenBSD Secure Shell server...
Apr 29 15:54:46 ubuntu systemd[1]: ssh.service: Deactivated successfully.
Apr 29 15:54:46 ubuntu systemd[1]: Stopped ssh.service - OpenBSD Secure Shell server.
Apr 29 15:54:46 ubuntu systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Apr 29 15:54:46 ubuntu sshd[2819]: Server listening on 0.0.0.0 port 22.
Apr 29 15:54:46 ubuntu sshd[2819]: Server listening on :: port 22.
Apr 29 15:54:46 ubuntu systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Apr 29 16:37:36 ubuntu sshd[2819]: Received signal 15; terminating.
Apr 29 16:37:36 ubuntu systemd[1]: Stopping ssh.service - OpenBSD Secure Shell server...
Apr 29 16:37:36 ubuntu systemd[1]: ssh.service: Deactivated successfully.
Apr 29 16:37:36 ubuntu systemd[1]: Stopped ssh.service - OpenBSD Secure Shell server.
-- Boot b38845cf582349d945b17d16d1e927f --
May 02 04:51:41 ubuntu systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
May 02 04:51:41 ubuntu sshd[1275]: Server listening on 0.0.0.0 port 22.
May 02 04:51:41 ubuntu systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
May 02 04:51:41 ubuntu sshd[1275]: Server listening on :: port 22.
ubuntu@ubuntu:~$

```

Figure 37: Viewing SSH logs using journalctl without pagination

The `journalctl -u ssh --no-pager` command was used to display SSH service logs directly in the terminal without pagination. The output shows detailed events such as service start, stop, and restart operations, allowing effective monitoring of SSH activity.



```

May 2 02:43
ubuntu@ubuntu: ~
May 02 04:51:41 ubuntu sshd[1275]: Server listening on 0.0.0.0 port 22.
May 02 04:51:41 ubuntu systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
May 02 04:51:41 ubuntu sshd[1275]: Server listening on :: port 22.
ubuntu@ubuntu:~$ sudo cat /var/log/auth.log | grep ssh
2026-04-29T15:23:48.934647+00:00 ubuntu sudo: ubuntu : TTY=pts/0 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/apt
install openssh-server
2026-04-29T15:24:20.868149+00:00 ubuntu useradd[3455]: new user: name=sshd, UID=122, GID=65534, home=/run/sshd, shell=/u
r/sbin/nologin, from=none
2026-04-29T15:24:54.104860+00:00 ubuntu sudo: ubuntu : TTY=pts/0 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/syst
emctl enable ssh
2026-04-29T15:25:15.732295+00:00 ubuntu sudo: ubuntu : TTY=pts/0 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/syst
emctl start ssh
2026-04-29T15:25:15.936967+00:00 ubuntu sshd[4313]: Server listening on 0.0.0.0 port 22.
2026-04-29T15:25:15.937512+00:00 ubuntu sshd[4313]: Server listening on :: port 22.
2026-04-29T15:25:24.694378+00:00 ubuntu sudo: ubuntu : TTY=pts/0 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/syst
emctl status ssh
2026-04-29T15:27:19.880122+00:00 ubuntu sudo: ubuntu : TTY=pts/0 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/nano
/etc/ssh/sshd_config
2026-04-29T15:53:15.499095+00:00 ubuntu sshd[1158]: Server listening on 0.0.0.0 port 22.
2026-04-29T15:53:15.499443+00:00 ubuntu sshd[1158]: Server listening on :: port 22.
2026-04-29T15:54:19.882306+00:00 ubuntu sudo: ubuntu : TTY=pts/0 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/nano
/etc/ssh/sshd_config
2026-04-29T15:54:46.416394+00:00 ubuntu sudo: ubuntu : TTY=pts/0 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/syst
emctl restart ssh
2026-04-29T15:54:46.444445+00:00 ubuntu sshd[1158]: Received signal 15; terminating.
2026-04-29T15:54:46.524233+00:00 ubuntu sshd[2819]: Server listening on 0.0.0.0 port 22.
2026-04-29T15:54:46.524766+00:00 ubuntu sshd[2819]: Server listening on :: port 22.
2026-04-29T15:56:50.624182+00:00 ubuntu sudo: ubuntu : TTY=pts/0 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/apt
install openssh-client
2026-05-02T04:51:41.470065+00:00 ubuntu sshd[1275]: Server listening on 0.0.0.0 port 22.
2026-05-02T04:51:41.470355+00:00 ubuntu sshd[1275]: Server listening on :: port 22.
2026-05-02T02:07:08.673321+00:00 ubuntu sudo: ubuntu : TTY=pts/0 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/fire
wall cmd --add-service=ssh --permanent

```

Figure 38: Filtering SSH logs using grep

The `cat /var/log/auth.log | grep ssh` command was used to filter SSH-related log entries

from the authentication log file. This approach helps in isolating relevant events such as login attempts, service activity, and user actions for easier analysis.

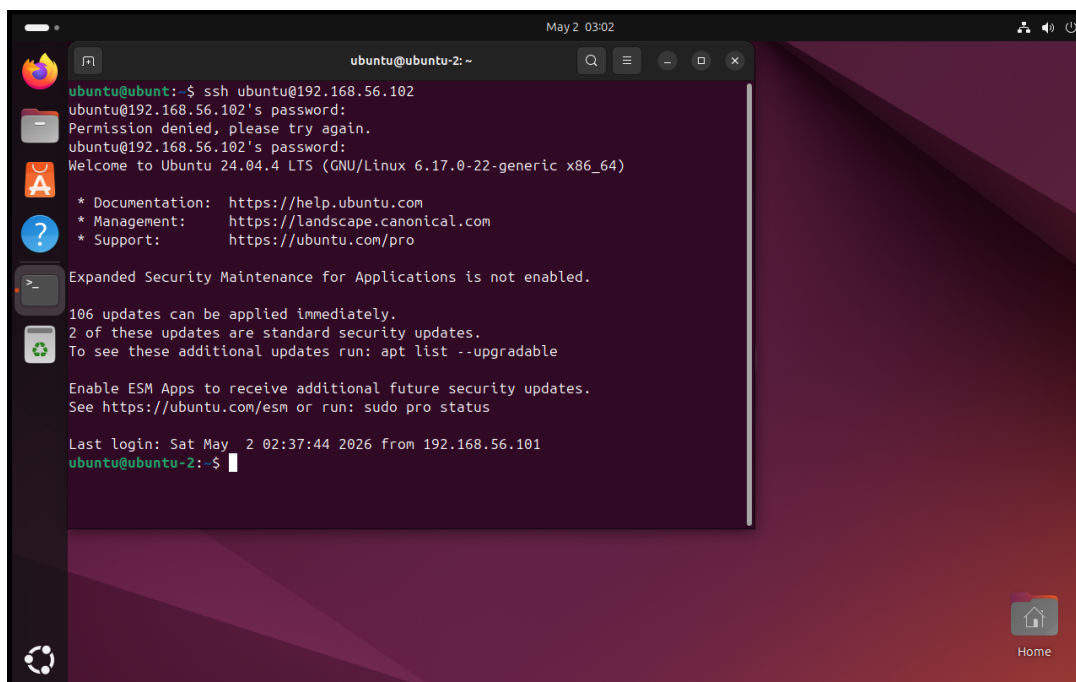


Figure 39: Failed and successful SSH login attempts

An SSH login attempt was performed where an incorrect password was entered initially, resulting in a failed authentication message. A subsequent attempt with the correct password successfully established the SSH session. This demonstrates how authentication failures and successful logins can be observed during SSH access.

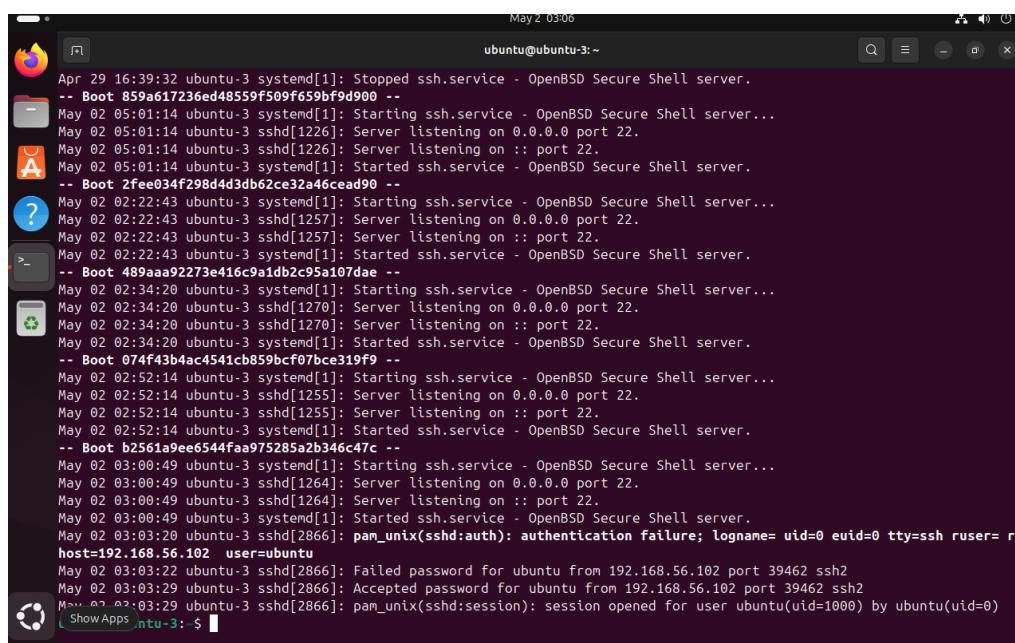
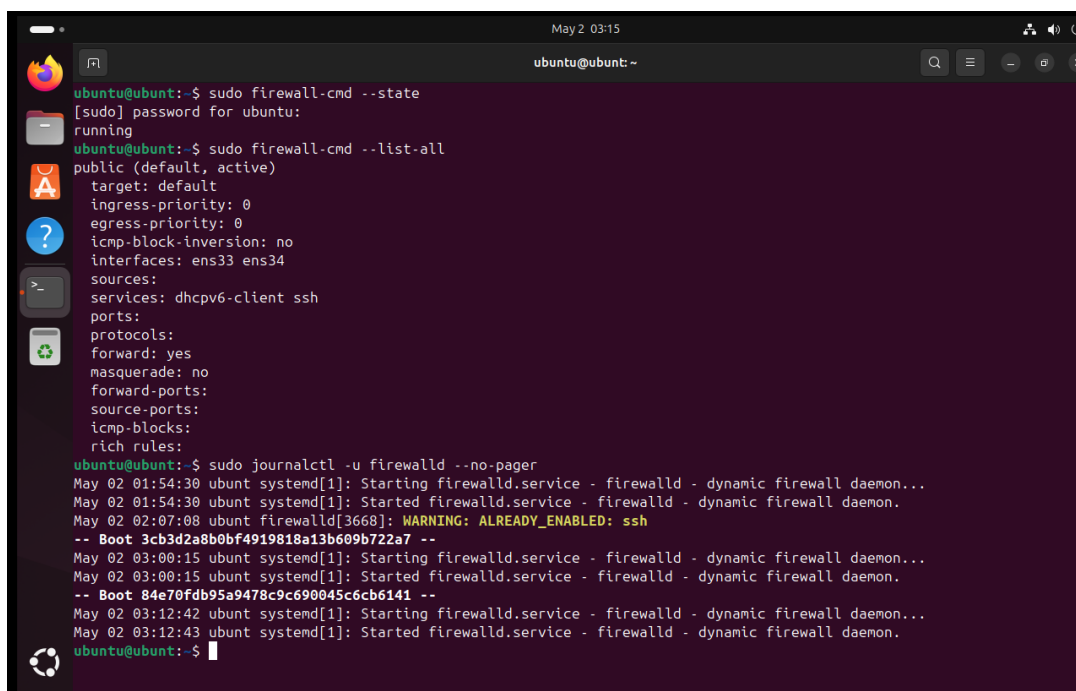


Figure 40: SSH login attempts (failed and successful)

SSH logs were analyzed to monitor authentication activities on the system. The logs display

multiple SSH events, including service start operations and login attempts.



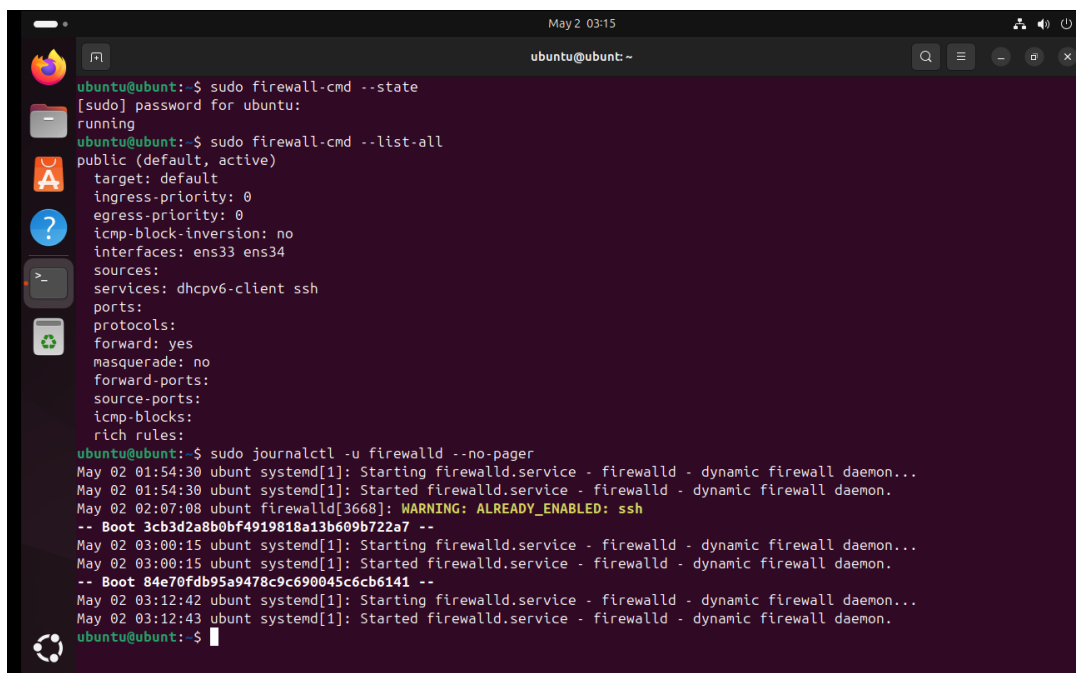
```

ubuntu@ubuntu:~$ sudo firewall-cmd --state
[sudo] password for ubuntu:
running
ubuntu@ubuntu:~$ sudo firewall-cmd --list-all
public (default, active)
  target: default
  ingress-priority: 0
  egress-priority: 0
  icmp-block-inversion: no
  interfaces: ens33 ens34
  sources:
  services: dhcpv6-client ssh
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
ubuntu@ubuntu:~$ sudo journalctl -u firewalld --no-pager
May 02 01:54:30 ubuntu systemd[1]: Starting firewalld.service - firewalld - dynamic firewall daemon...
May 02 01:54:30 ubuntu systemd[1]: Started firewalld.service - firewalld - dynamic firewall daemon.
May 02 02:07:08 ubuntu firewalld[3668]: WARNING: ALREADY_ENABLED: ssh
-- Boot 3cb3d2a8b0bf4919818a13b609b722a7 --
May 02 03:00:15 ubuntu systemd[1]: Starting firewalld.service - firewalld - dynamic firewall daemon...
May 02 03:00:15 ubuntu systemd[1]: Started firewalld.service - firewalld - dynamic firewall daemon.
-- Boot 84e70fdb95a9478c9c690045c6cb6141 --
May 02 03:12:42 ubuntu systemd[1]: Starting firewalld.service - firewalld - dynamic firewall daemon...
May 02 03:12:43 ubuntu systemd[1]: Started firewalld.service - firewalld - dynamic firewall daemon.
ubuntu@ubuntu:~$

```

Figure 41: Verifying firewall status, rules, and logs

The firewall status was verified using the ‘firewall-cmd --state’ command, confirming that the service is running. The active firewall configuration was displayed using ‘--list-all’, showing that the SSH service is allowed. Additionally, firewall logs were examined using ‘journalctl -u firewalld --no-pager’, confirming firewall activity and rule enforcement.



```

ubuntu@ubuntu:~$ sudo firewall-cmd --state
[sudo] password for ubuntu:
running
ubuntu@ubuntu:~$ sudo firewall-cmd --list-all
public (default, active)
  target: default
  ingress-priority: 0
  egress-priority: 0
  icmp-block-inversion: no
  interfaces: ens33 ens34
  sources:
  services: dhcpv6-client ssh
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
ubuntu@ubuntu:~$ sudo journalctl -u firewalld --no-pager
May 02 01:54:30 ubuntu systemd[1]: Starting firewalld.service - firewalld - dynamic firewall daemon...
May 02 01:54:30 ubuntu systemd[1]: Started firewalld.service - firewalld - dynamic firewall daemon.
May 02 02:07:08 ubuntu firewalld[3668]: WARNING: ALREADY_ENABLED: ssh
-- Boot 3cb3d2a8b0bf4919818a13b609b722a7 --
May 02 03:00:15 ubuntu systemd[1]: Starting firewalld.service - firewalld - dynamic firewall daemon...
May 02 03:00:15 ubuntu systemd[1]: Started firewalld.service - firewalld - dynamic firewall daemon.
-- Boot 84e70fdb95a9478c9c690045c6cb6141 --
May 02 03:12:42 ubuntu systemd[1]: Starting firewalld.service - firewalld - dynamic firewall daemon...
May 02 03:12:43 ubuntu systemd[1]: Started firewalld.service - firewalld - dynamic firewall daemon.
ubuntu@ubuntu:~$

```

Figure 42: Firewall status verification and log inspection

The firewall service was verified to be running using the ‘firewall-cmd --state’ command. The

active firewall configuration was displayed using ‘`firewall-cmd --list-all`’, confirming that the SSH service is enabled. Furthermore, firewall activity logs were examined using ‘`journalctl -u firewalld --no-pager`’, demonstrating that the firewall service is actively running and logging events.

Part E: Task 13: Short Analysis & Reflection

Logs play a critical role in detecting unauthorized access attempts as they record all system activities, including both successful and failed SSH login attempts. By analyzing these logs, administrators can identify suspicious behaviors such as repeated failed login attempts, which may indicate brute-force attacks. Continuous monitoring is essential in responsible network management because it enables early detection of abnormal or malicious activities. This allows administrators to respond quickly and mitigate potential threats before they escalate. Additionally, logging and monitoring help enforce access control policies by ensuring that only authorized users can access system services. As a result, these practices protect system resources from misuse and safeguard sensitive user data. Overall, they contribute significantly to maintaining a secure and reliable network environment.

8 Part F: Network Traffic Analysis with Wireshark

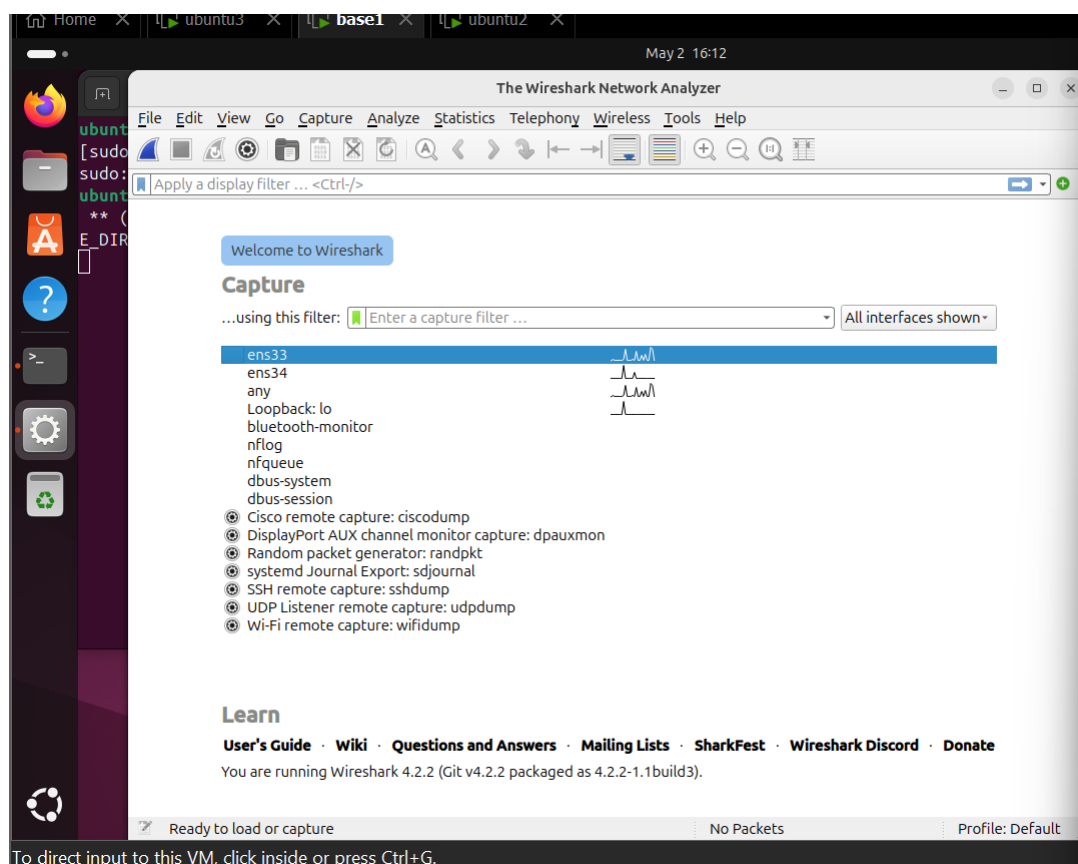


Figure 43: Launching Wireshark and selecting a network interface

Wireshark was launched on the Ubuntu Base machine, and the available network interfaces were displayed. The active interface (`ens33`) was selected to begin capturing network traffic for

further analysis.

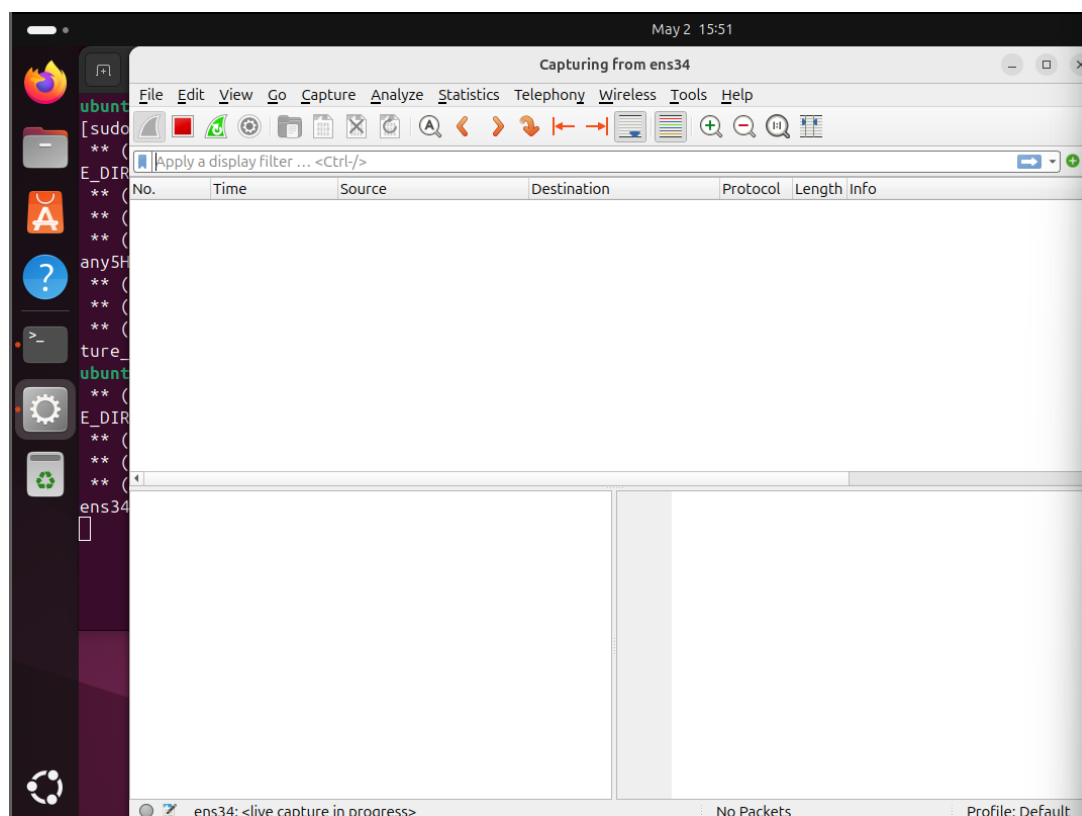


Figure 44: Starting packet capture before generating network traffic

Packet capture was initiated on the selected network interface using Wireshark. At this stage, no traffic has been generated yet, indicating that the capture process is ready to monitor upcoming network activities such as ping and SSH communications.

Observation: Minimal traffic is observed before executing ping, indicating that no significant communication had started at this stage.

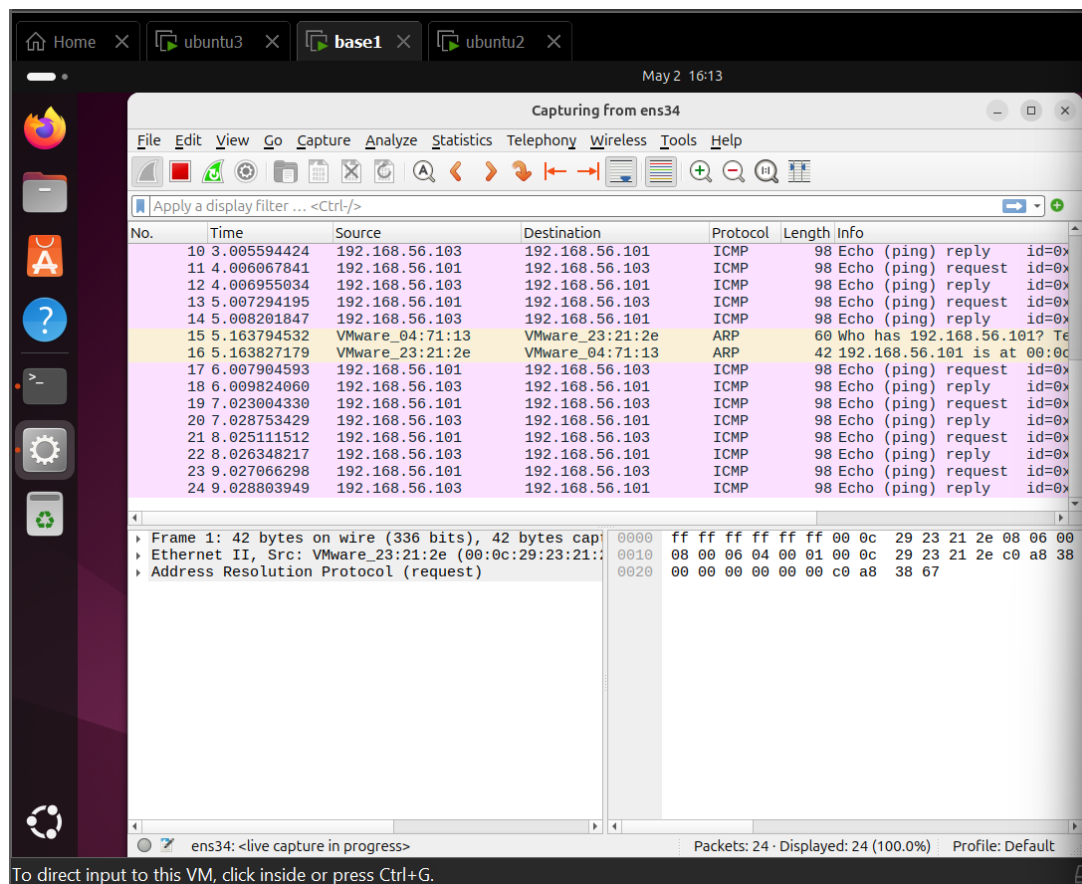


Figure 45: Capturing ICMP packets after executing ping

After initiating the packet capture, ICMP traffic was generated using the ping command. Wireshark successfully captured multiple ICMP echo requests and replies between the source and destination hosts, confirming active network communication and proper packet transmission.

Observation: The captured packets show ICMP echo requests and replies between the source and destination IP addresses, indicating successful communication with no packet loss.

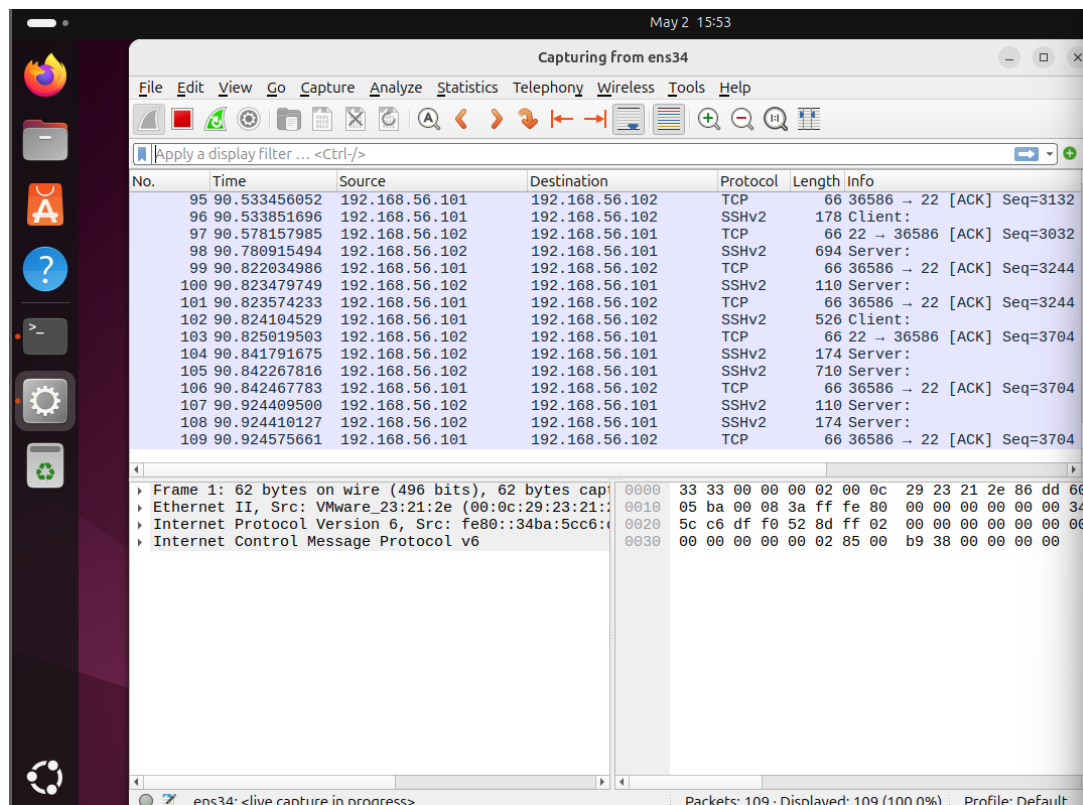


Figure 46: Capturing SSH traffic using Wireshark

Wireshark was used to capture and analyze SSH communication between hosts. The captured packets show TCP segments associated with SSH (port 22) along with SSHv2 protocol exchanges between the client and server. This confirms that SSH traffic is successfully transmitted over the network.

Observation: The traffic appears encrypted and uses port 22, confirming secure communication between the client and server using the SSH protocol.

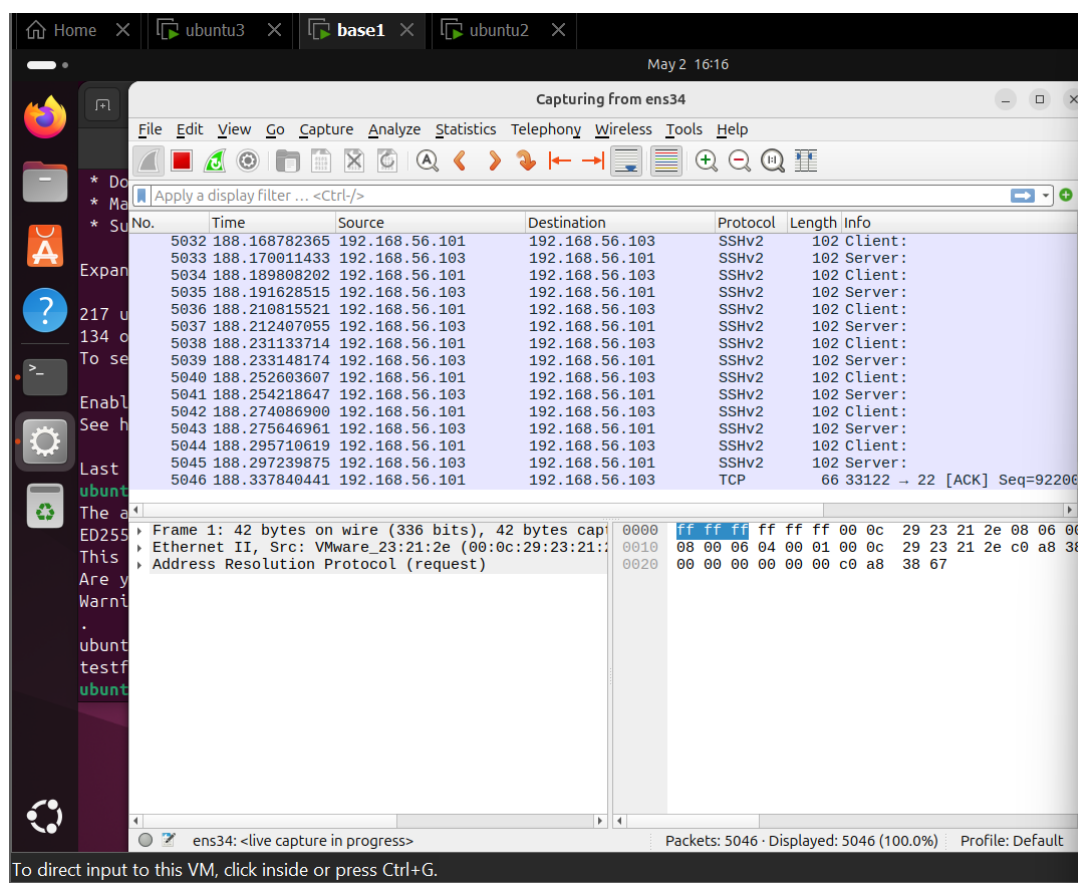


Figure 47: Capturing SCP file transfer traffic in Wireshark

A file transfer was performed using the SCP protocol between hosts. Wireshark captured the corresponding network traffic, showing TCP segments and SSHv2 packets exchanged between the source and destination. This demonstrates that SCP file transfer operates over the SSH protocol using TCP port 22.

Observation: The captured traffic shows data transfer over TCP using SSH, confirming that the file was securely transmitted between the machines.

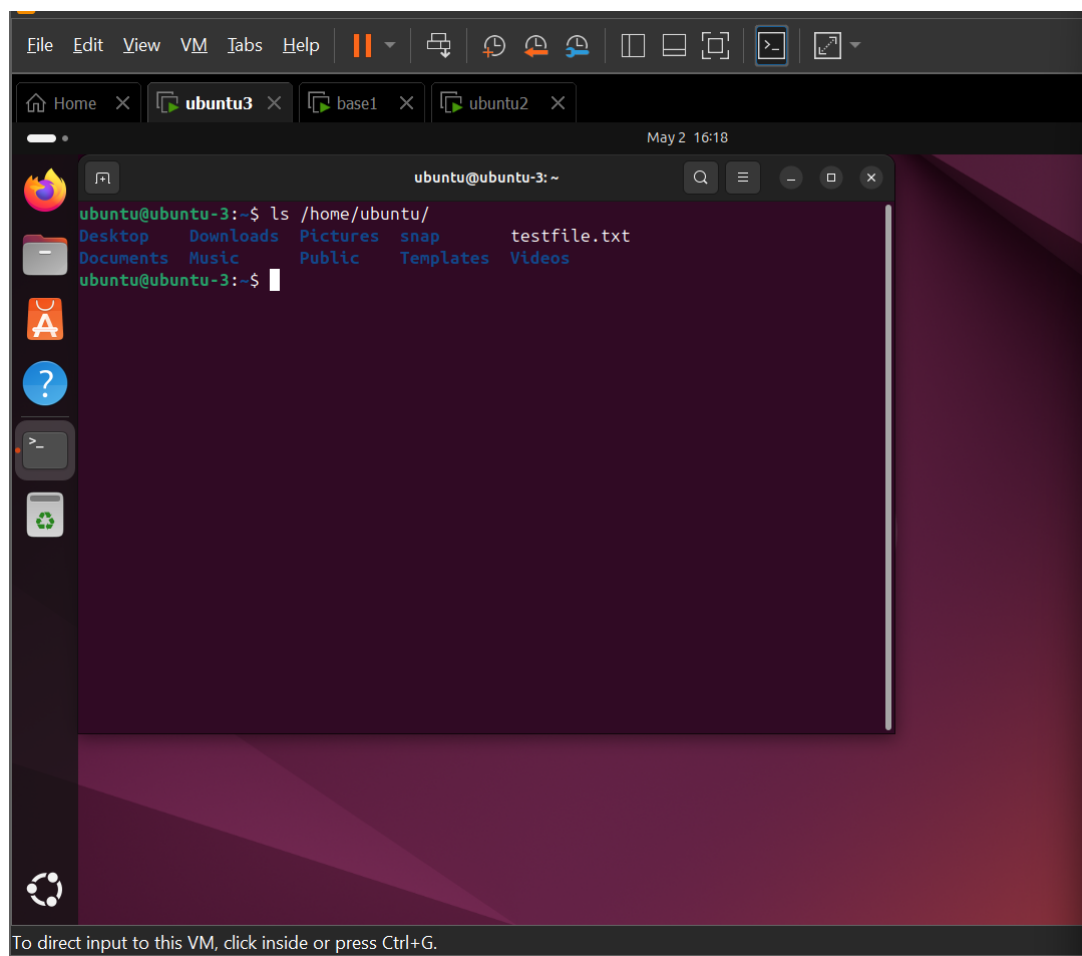


Figure 48: Verifying file transfer on the destination machine

After transferring the file using SCP, the destination machine was accessed and the contents of the home directory were listed using the 'ls' command. The presence of 'testfile.txt' confirms that the file was successfully transferred and stored on the target system.

Observation: The presence of the file in the destination directory confirms that the transfer process was completed successfully.

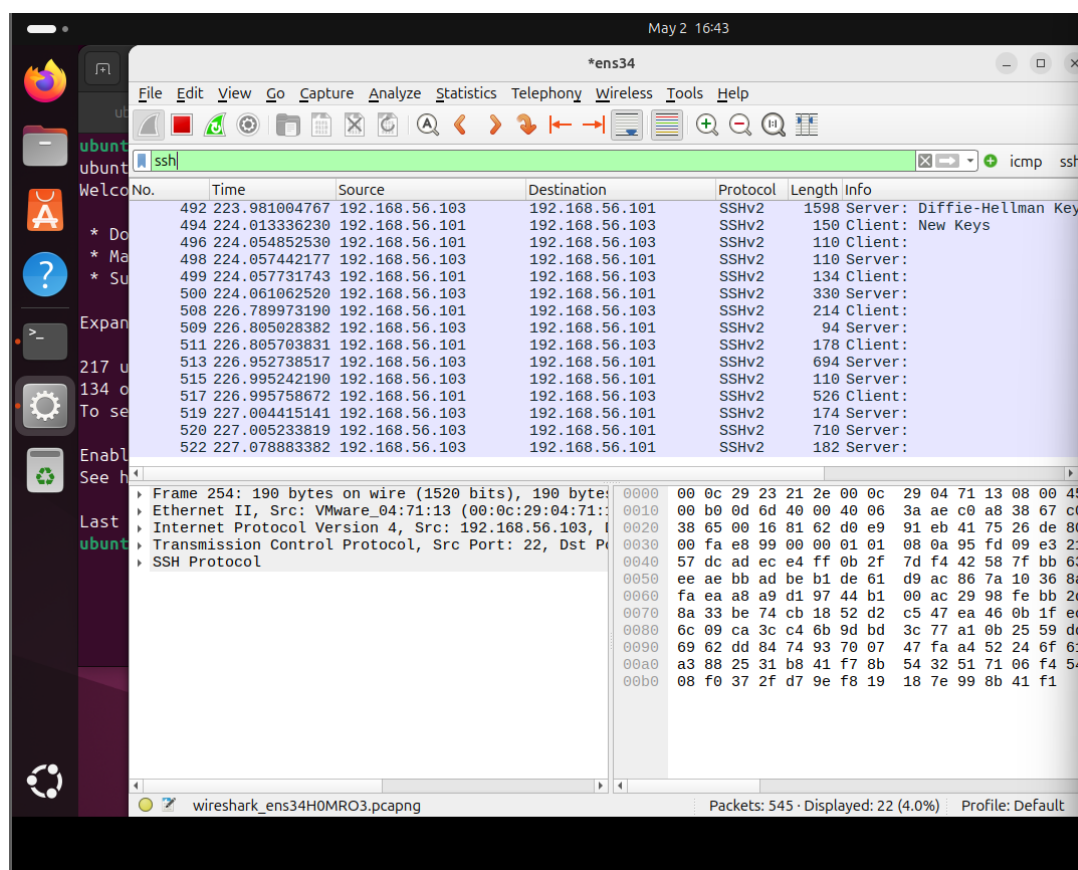


Figure 49: Applying SSH display filter in Wireshark

A display filter ('ssh') was applied in Wireshark to isolate SSH traffic from other network packets. The filtered results show SSHv2 protocol exchanges between the communicating hosts, allowing focused analysis of secure shell communication and related data transmission.

Observation: Only SSH packets are displayed after applying the filter, making it easier to analyze encrypted communication between devices.

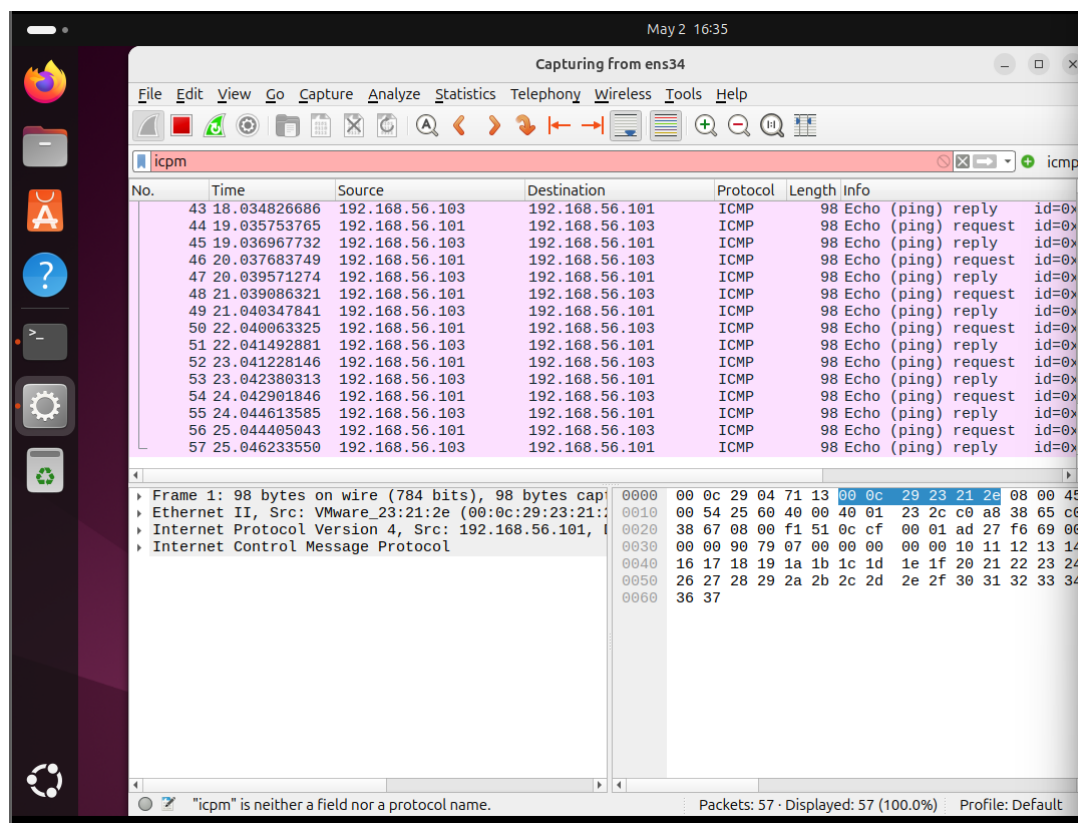


Figure 50: Applying ICMP display filter in Wireshark

A display filter ('icmp') was applied in Wireshark to isolate ICMP packets generated during the ping process. The captured packets show echo request and echo reply messages exchanged between the hosts, confirming successful connectivity and allowing detailed inspection of ICMP communication.

Observation: The filter isolates ICMP packets, clearly showing echo requests and replies exchanged between the hosts.

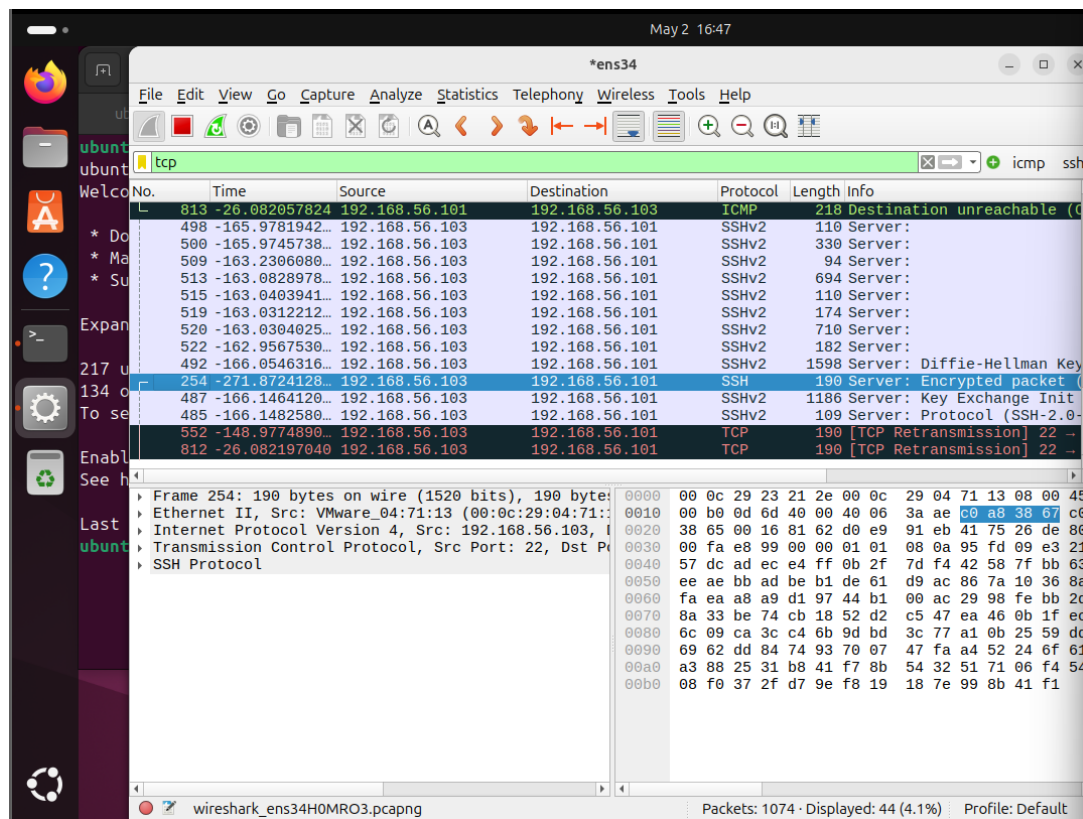


Figure 51: Applying TCP display filter in Wireshark

A display filter ('tcp') was applied in Wireshark to isolate TCP traffic during network communication. The captured packets include SSH-related exchanges over TCP port 22, along with retransmissions and encrypted packets. This confirms that secure communication is established over TCP and allows detailed analysis of transport-layer behavior.

Observation: The TCP filter highlights transport-layer communication, including SSH sessions and data transfer behavior.